I-Scan® Equilibration and Calibration Practical Suggestions.

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INTRODUCTION

This guide outlines the issues supporting the practical operation of ***I-Scan*** and array sensors, including Equilibration and Calibration. It is not meant to be taken as a set of rules or a step-by-step listing on how to perform these functions for any one specific application, but is intended to answer to some common questions. This guide will give a more in-depth discussion of the physics behind sensor performance. Where applicable, examples are provided and diagrams displayed in order to illustrate user experience, as well as our own Research and Development findings.

This document provides an overview and describes steps that are generally beneficial. As you gain experience with a particular application, you may find reasons to deviate from this guide. Tekscan welcomes your comments and suggestions that may result in refinements or revisions to this material. As we learn from you, we will share the results with other users.

Note: This document assumes you are familiar with the User Manual that was provided with your sensor system, as well as a familiarity with the proper use of the sensor and equipment.

OVERVIEW

When approaching a new application, the recommended order of operation to achieve the best performance is the following:

1. **Select a sensor with the appropriate saturation pressure**. Estimate the saturation pressure needed. Then perform a quick, trial experiment with a sensor to determine if the sensor selected has the right physical size and saturation pressure for the expected pressures. This will validate your sensor selection.

2. **Condition the sensor**. Load the sensor so the initial change of sensitivity occurs before serious measurement begins.

3. **Equilibrate/Normalize the sensor**. Confirm that each sensel has a uniform output from a uniform load, and correct for uneven output. If equilibration is used, it is important to equilibrate before calibrating.

4. **Calibrate the sensor**. Mimic the experiment for a good calibration by placing objects with similar surface characteristics to those that will be used in the experiment on the sensor with a similar known force or load, for a similar duration.

5. **Record a movie**. Use a frame rate fast enough to capture dynamic behavior of interest. Because data is taken from the sensels sequentially, in row and column fashion, a scan rate about five to ten times faster than the frequency of the event is needed to get representative data. Triggering may be used to start the recording when an interesting event is expected.

6. **Edit the movie.** Editing can be used to remove unwanted data, and cut frames that are not relevant.

Note: Each of the steps above are optional. Which steps are performed will depend primarily on the application and goals of the experiment.

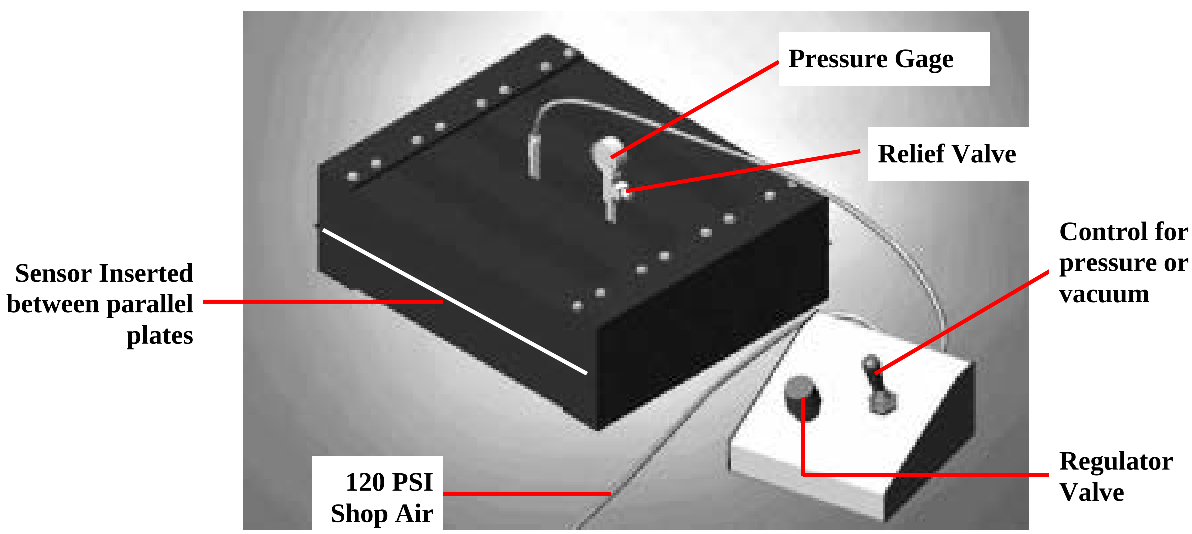
PRECONDITIONING THE SENSOR

Before using a sensor, precondition the sensor. It is preferable to use a load greater than the test load, however, if you are unable to do so, conditioning the sensor with any amount of load is still better than not conditioning the sensor at all. Preconditioning the sensor pad minimizes drift, hysteresis, and improves repeatability during the experiment. It is therefore recommended to precondition the sensor before conducting experiments or recording data.

Tekscan's equilibrator unit may be used for pre-conditioning since it imparts a uniform load pneumatically. If possible, it helps to have a similar temperature during the preconditioning and the experiment. Use the equilibrator if the pressure expected is lower than the equilibrator's capacity. For higher pressures, you may want to use two metal plates in a press with a thin sheet of urethane foam on the sensor. The foam should be compliant enough to accommodate differences in surfaces coming together and have a low Poisson's Ratio so the foam will not "squirm" markedly on the sensor surface.

Tekscan offers several equilibrators (most are pneumatic and one is electric / hydraulic). These equilibrators have an internal urethane bladder that fills with air or liquid to apply a uniform pressure to the sensor pad. An analog pressure gauge, external pressure regulator (three-position toggle switch), and plastic tubing are included as well as a pressure relief valve. Some models are constructed from wood and some from aluminum.

Example of a Tekscan Equilibrator



In some cases, the sensors may stick in the equilibrator, or may not slide smoothly in and out of it. This effect may be more pronounced on humid days. A very light dusting of talcum powder on the sensor surface makes it easier to slide. Talcum powder works better than cornstarch, because it is less hygroscopic. Rather than spreading the talcum powder with your hand, consider putting some of it on a cotton rag or piece of gauze. This avoids getting finger oil on the sensor surface. If the sensor is vented, try to avoid talcum powder near the vents. Foreign material sucked into the sensor will cause spurious output.

When using an equilibrator with thick plates, such as the PB100B, the height of the slot where the sensor exits the equilibrator is much higher than the table on which the equilibrator is placed. Consider using a cardboard box or spacer to support the handle at the height of the slot. This reduces strain on the neck of the sensor, and reduces the amount by which the neck is flexed during equilibration.

EQUILIBRATION (NORMALIZATION)

Overview

Equilibration involves applying a uniform pressure to the entire active area of the sensor. Then ***I-Scan*** software determines a gain (scale factor) for each sensel such that its Digital Output (**DO**) is equal to the average **DO** of all loaded sensels. The average **DO** is not affected by equilibration. Sensels with low initial output get more gain to raise their **DO**, and those with high initial output get reduced gain to lower their **DO**. The result is digital compensation for each individual sensel. This compensates for differences in sensitivity between sensels due to manufacturing, or due to repeated use of the sensor. Applying a uniform pressure is also useful as a quality assurance check on the sensor, to confirm that the sensor has an acceptable uniform output, is operating correctly, and no rows or columns have lost continuity.

In general, a sensor's output (sensitivity) will exhibit a decrease when it is loaded repeatedly. This decrease occurs during use and is a function of the loading conditions. A good application for equilibration is when the sensor is loaded repeatedly in the same physical location on the sensor pad. Typically, the region of the sensor pad that has not been loaded retains its original sensitivity, that is, it maintains a higher sensitivity than the loaded region. Equilibration compensates for the sensor's trends, thus restoring an even sensitivity within the sensor. This, in turn, extends the useful life of the sensor.

Single-load Equilibration should be performed at a pressure typical of, or important for, the test. Multi-load equilibration is preferred as it gives the assurance of uniform sensel behavior over a larger range of pressures. If there is only one equilibration point, it should be in the midst of the pressures of interest.

Equilibration may be performed well in advance of the measurement, and stored as a file. The equilibration file may be loaded into the real-time window, so only equilibrated data is displayed. After the sensor has had significant experience, or when the user wants to reconfirm sensor performance, equilibration can, and should be repeated.

Some other points to consider regarding Equilibration (Normalization) include the following:

- To get uniform pressure distribution, it is usually necessary to use air or liquid under pressure in a bladder.

- A less accurate equilibrator can be built with a layer of compliant material, backed up with a stiff plate and a means to apply load to the plate.

- Equilibration should be performed before calibration.

- An equilibration file may be loaded before or after the new recording. *Note: this depends on the version of software being run. Some versions do not allow equilibration after the recording is completed.*

- Single-Load Equilibration scale factors range from 0.5 to 2, and the scale factor is unique to each sensel. With a Multi-Load Equilibration, the scale factors range from 0.5 to 2, and the scale factor is not only unique to each sensel, but also to each equilibration pressure level (each equilibration point).

- The Black and White image in the equilibration dialog window represents the following:
    - cold (sensels are not as sensitive) with darker colors
    - hot (sensels are more sensitive) with lighter gray colors

- The sensels with light gray color receive equilibration factor values close to 0.5

- The sensels with darker color receive equilibration factor values close to 2.0

- During equilibration, ensure that there is not trapped air inside the sensor, which will give a “pillow effect”. Sometimes trapped air will slowly bleed out of a sensor in the equilibrator. In other cases, the substrate may be pierced to allow a path for air to escape. Refer to “Low Pressure Application” for further information.

- Sensors may be sequentially loaded a portion at a time in order to “purge” any air out from inside the sensor. When 75% of a sensor is inserted into the equilibrator, the trapped air is squeezed into the 25% of the sensor that is not loaded.

Single-Load Equilibration

Single-Load Equilibration normalizes the sensor using a single pressure value; typically a pressure of interest for the test to be performed. The raw digital output of all sensels is averaged. Then the gain for any sensels higher than the average is reduced to produce the average output, and the gain for any sensels lower than the average is increased to produce the average output. Single-Load Equilibration is a quick, simple solution.

Multi-Load Equilibration

Multi-Load Equilibration normalizes the sensor at several different pressure points. It offers the advantage of compensating for sensel output over the full operating pressure range. A piece-wise linear set of correction factors is implemented to normalize raw digital data on a sensel by sensel basis. Multi-point equilibration treats each sensel individually in each band of Raw Digital Output. For each sensel, a series of lines are generated with slopes that determine the incremental increase of raw digital output in each band.

Earlier versions of ***I-Scan*** provided one pressure at which the sensor could be equilibrated. Multiple-load equilibration was implemented in ***I-Scan*** Version 4.23 and higher. The following outlines some of the features of Multiple-load equilibration:

- Tekscan offers up to ten equilibration points.
- A scale factor is determined at each “equilibration pressure” that brings the reported DO of each sensel to the average DO value of all loaded sensels.
- The actual digital value for each “equilibration pressure” is saved for each active sensel.
- An individual “piece-wise linear equilibration curve” is defined for each grid sensel.
- If a sensel is not active at an “equilibration pressure,” then the overall average digital value is used as its “actual” digital value, so it gets a scale factor of 1.
- The zero point is always a curve point. That is, there is no equilibration-induced zero offset.
- The highest two points are linearly extrapolated to define the line beyond the highest point.

Example

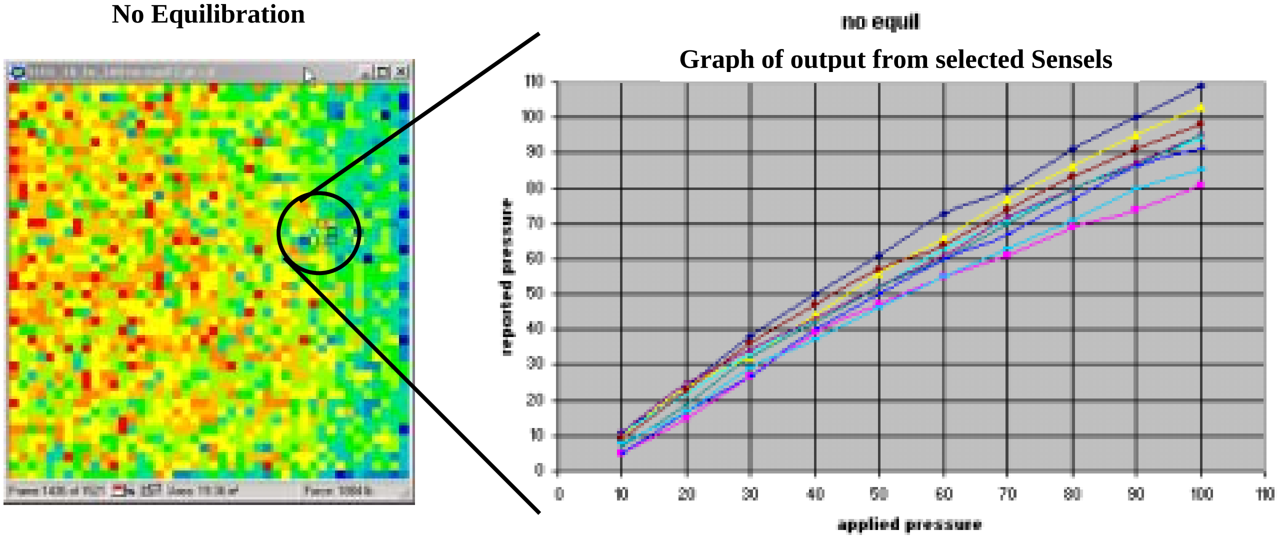
Application pressure range is 0 to 100 PSI. Multi-point equilibration is implemented with four pressures/loads:

20 PSI, 40 PSI, 60 PSI, 80 PSI, (Four-Load Equilibration)

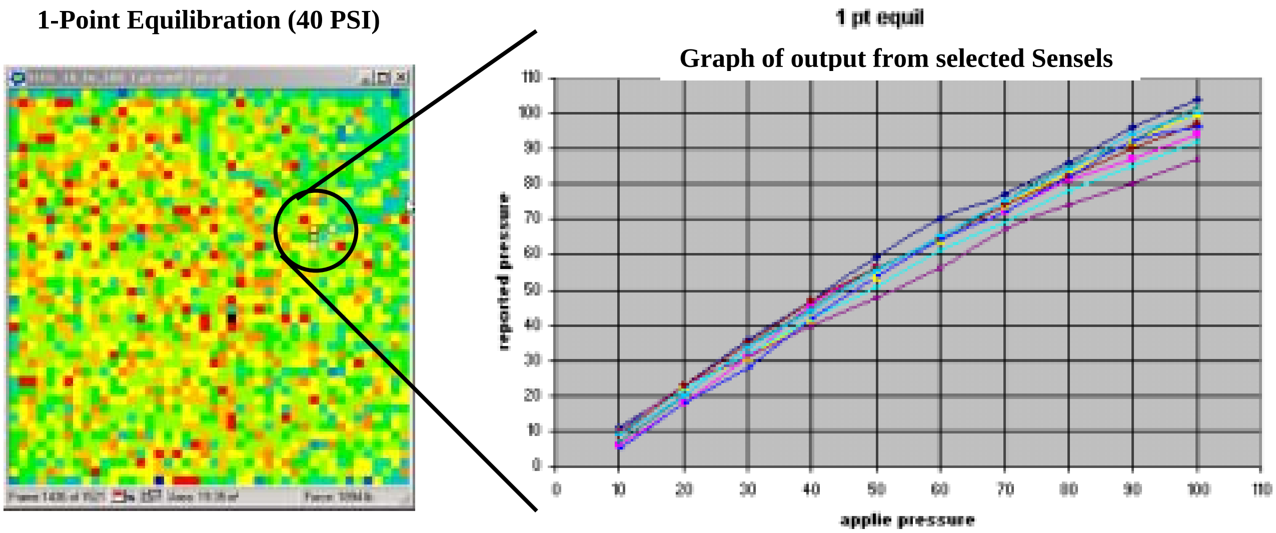
The following graphs outline the difference in using no equilibration, one-point equilibration, and four-point equilibration. Note the wider divergence in the sensels' reported pressure when no equilibration is used. One-point equilibration yields a tighter spread where the equilibration point is located, and four-point equilibration provides the tightest spread of the three graphs. The least divergence in the reported output of the sensels occurs where the equilibration points are located. As the pressure increases or decreases away from the equilibration point, this divergence gradually becomes greater.

The following set of images show the differences between single-load and multi-load equilibration:

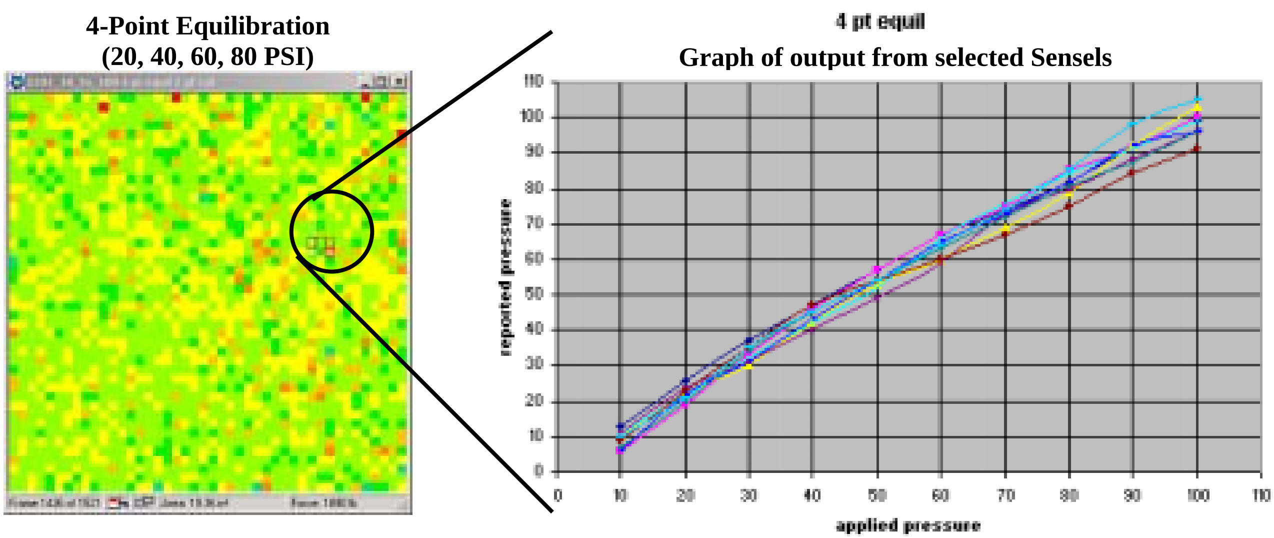
No Equilibration



1-Point Equilibration (40 PSI)



4-Point Equilibration
(20, 40, 60, 80 PSI)



CALIBRATION

Overview

Each sensel is a force sensitive variable resistor, whose impedance changes from above 10 Meg ohms (with no load) to less than 20 Kilo ohms (with full load). During Calibration, a force is applied to the sensor, thereby changing the impedance of the loaded sensels. The analog to digital converter assigns a digital value between 0 and 255 (8 bit) to each sensel, depending on its impedance value. Calibration is the process of correlating the digital output from the sensels to engineering units (Force or Pressure). Put another way, the DO (Digital Output) of the sensor is correlated to Pounds, Newtons, Kilograms, etc. (Force) and PSI, kPa, Kg/m^2, etc. (Pressure). Since error in calibration converts to error of the reported experimental load, care should be taken to properly calibrate the sensor. For example, if ***I-Scan*** is told that the calibration load is 30 pounds, when 35 pounds is actually applied, the reported loads and pressures will all be low, by about 15%.

Note: The reported calibrated data should be periodically compared to a known reference load to ensure the calibration is still within an acceptable range.

The overall system accuracy is generally +/- 10% of the full scale. Accuracy depends on the care taken by the user.

Single Load (linear) Calibration.

Single-Load Calibration assumes the sensor has zero output with zero applied load. The user applies a known force to get a single calibration point. ***I-Scan*** draws a straight line between the two points (zero point and the calibration point). The line is extrapolated to a digital output of 255, and saturation pressure is displayed.

For experimental loads in the vicinity of the calibration load, single point calibration gives accurate results. If the experimental load is small or large compared to the calibration load, errors will grow. Typically, the graph of sensor Digital Output from varying load is a smooth curved line following a power law curve. A straight line from the origin is close to the curve near the calibration point, but increasingly divergent at the extremes.

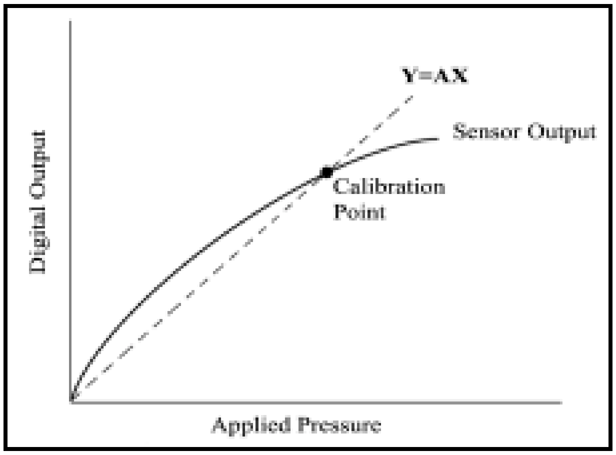
Single-Load Calibration uses the following equation:

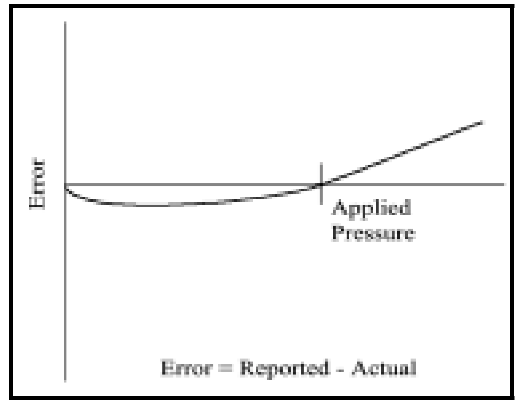
*Digital Output / Number of Sensels Loaded = (Global Scale Factor) * (Entered Force / Area Loaded)*

Yielding an equation of the form: **Y=AX**

Y = Force or Load (in engineering units)
A = Calibration Factor (determines slope)
X = Digital Output (may or may not be equilibrated).

The following 2 images display the single-load calibration concept, and error:





The following outlines some of the Advantages of a Single-Load (Linear) Calibration:

- It is easy and quick to perform.
- It is easy to understand.
- It is good for loads with small variation in output range over time.
- Since the response of Tekscan sensors is approximately linear, and since many experiments have other sources of error, single point calibration is preferred in many cases. This is especially true if successive experiments involve the same approximate force.
- Single point calibration is preferred when the *pressure distribution* with *different total loads* is similar. This happens when contact area increases with increasing load. Two examples are as follows:
  - **Tire studies.** When a tire contacts a surface, the inflation pressure of the tire determines the pressure. As the load is increased, the contact area grows, but the inflation pressure remains relatively constant, so the pressure distribution stays relatively constant.
  - **Seating studies.** For compliant surfaces coming together, such as the human body, increased load is associated with increased area. The sensel pressure distribution under a light person may be just about the same as the sensel pressure distribution under a heavy person. In other words, the contact area under a light person is usually small and the contact area under a heavy person is usually large.

Two-Load (Non-Linear) Calibration

Two-Load calibration uses the load distribution information, algebra and a numerical technique (iteration) to calculate the power law equation used for sensor calibration. For this reason, the applied loads must be precisely known and entered into the software correctly, and the calibration loads must generate different pressure distributions. Doing so will yield an accurate sensor calibration.

In Two-Load calibration, if the loads at the two calibration points are about the same, due to the algebra involved, large errors in the derived equation can result. As a rule of thumb, the Force1/Area1 should be 2 to 3 times different than the Force2/Area2.

Two-Load Calibration utilizes a sophisticated algorithm, which processes the distribution through a 'histogram' method. The two-point algorithm includes zero load to solve for the two constants in the exponential Power Law equation $\mathbf{Y = AX^b}$

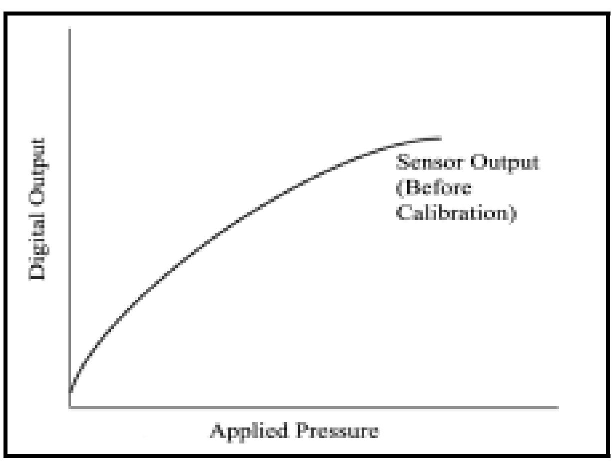
> **Y** = force or load (in engineering units)
> **A** = scale factor (determines slope)
> **X** = raw digital output
> **b** = exponent (determines curvature)

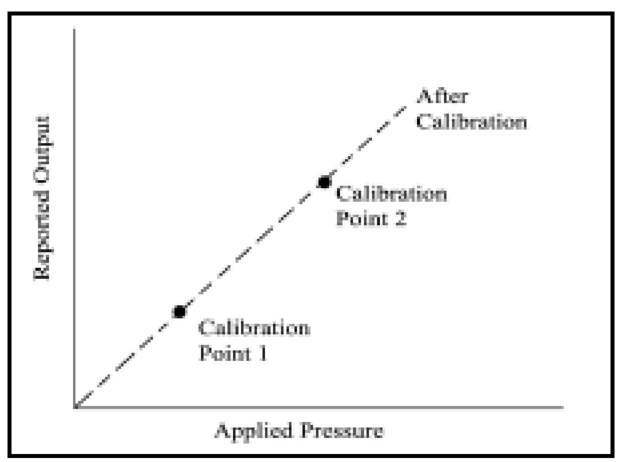
To illustrate two-load calibration, if the typical target load in the experiment is 1950 pounds, for the calibration apply a load of 390 pounds and take a calibration point. Then apply a load of 2600 pounds to get a second calibration point. This will give a power law curve of the form $Y = AX^b$. If the sensor is perfectly linear, the value of b will be 1. The software allows values of b between 0.1 and 4. Please be suspicious of values for b less than 0.6 or greater than 2.

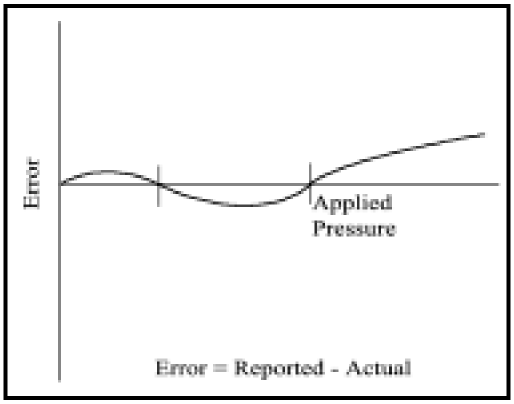
Note: to view the value of "b", after doing a two-load calibration, click on Options → Setting → Calibration.

When looking at the calibration window, for two-load calibration, the calibration line may or may not pass through the points of DO and calculated result. The points displayed reflect average loads. Depending on the statistics of load on each sensel, the algebra determines a calibration curve.
The points represent the average output and depending on the skew of the load distribution histogram, the curve may be higher or lower than the calibration points.

The following images display the 2-load calibration concept, and expected error:



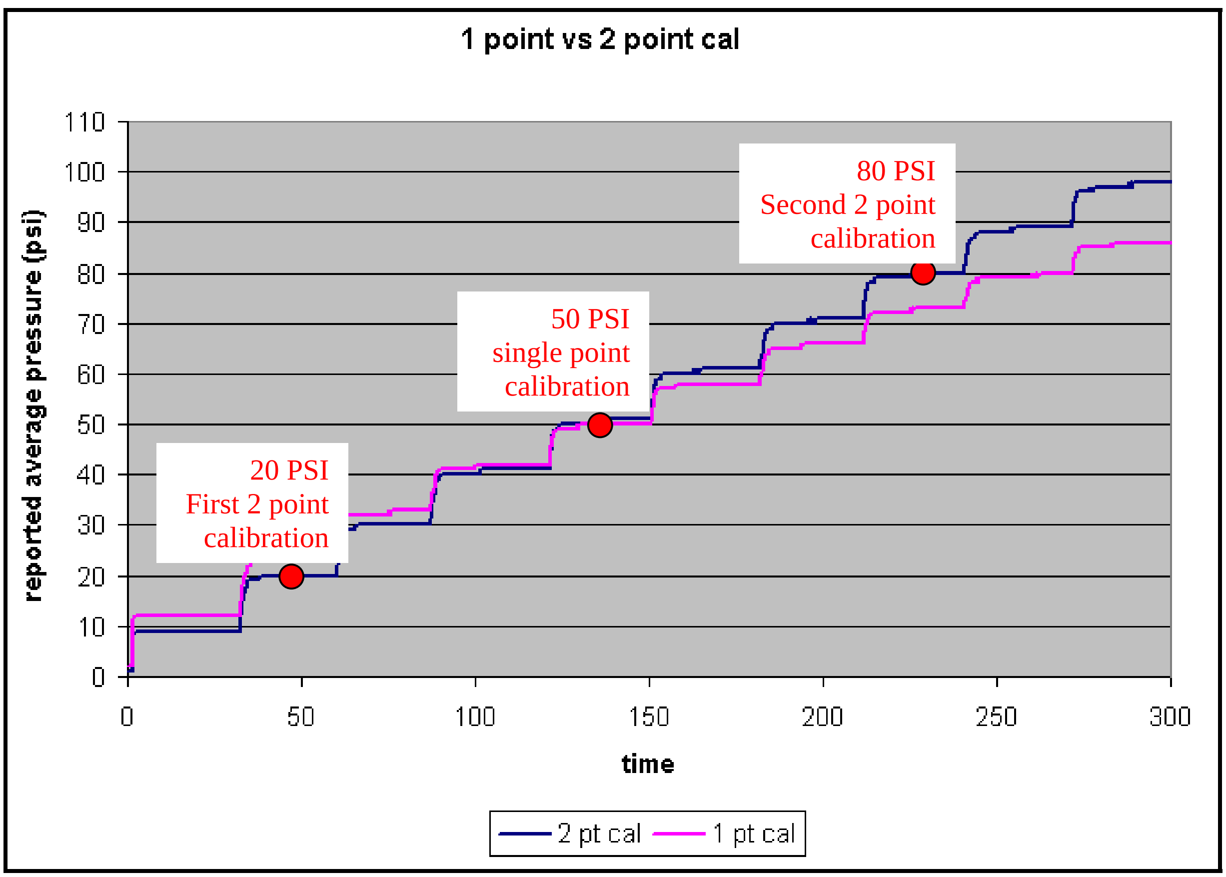




The advantage of a two-load (non-linear) calibration is that it may help *reduce errors*, and provide *greater accuracy*, especially if the total load in the experiment varies over a large range.

The following graph shows a single-load versus 2-load calibration. The single-load calibration was performed at approximately 50 PSI, while the 2-load calibration was done both at 20 PSI and 80 PSI. Notice that both sensors are equal at 50 PSI, where they both intersect on the graph.

This experiment was done in an equilibrator. Pressure was applied for 30 seconds in 10 PSI increments from 10 to 100 PSI. Note that with 10, 20, and 30 PSI applied, the single-load calibration case reads high, and with 60, 70, 80, and 90 PSI applied, it reads low. By comparison, the two-load calibration case gives low error throughout the range of applied pressure.



Alternative Calibration Approaches

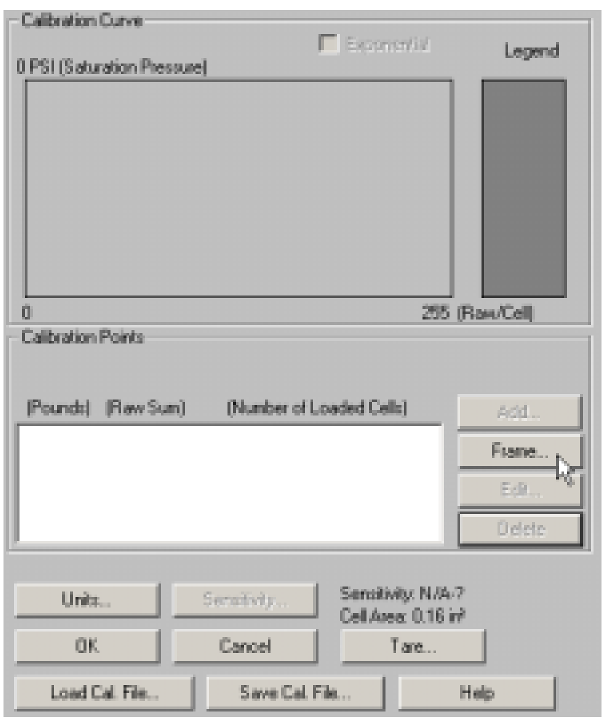
Frame (Post) Calibration

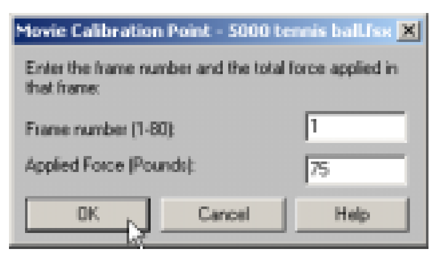
With Frame Calibration (also called Post Calibration), a user can calibrate the sensor after the data is recorded. The user must know the force acting through the sensor for at least one frame in order to use frame calibration.

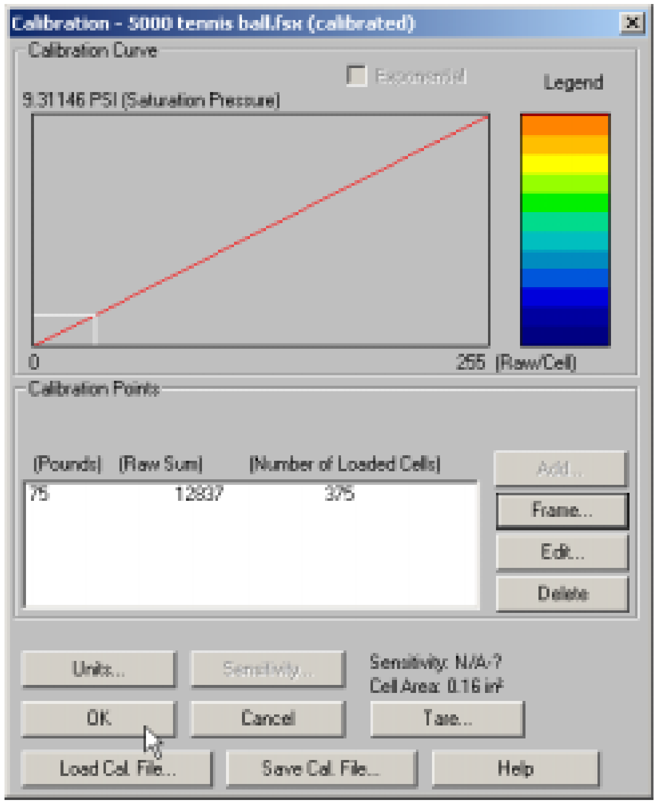
Use similar material interface, temperature, and timing if possible on a load frame (MTS / Instron, Tinious Olsen, etc.) for imparting the calibration force. Your options include:

- One-point calibration at a high load
- One-point calibration at a low load
- Two-point calibration, first using a representative frame with a known low force, then using a second representative frame with a known high force. For good results, $F_2 > 2F_1$, and $F_2/A_2 > 2(F_1/A_1.)$

The following images review the process of frame calibration. To access this dialog box, go into the "Tools" menu and select **Calibration**.







An example of frame calibration is when the experiment is done on a load machine, such as an Instron, or MTS. Then a frame typical of a point of interest in the movie where total applied force is known is used to calibrate the entire movie.

As a general rule for best calibration, the calibration event should mimic the experiment. For more information, refer to "Application Parameters to Mimic for Calibration".

Loading Calibration after Recording

Movie calibration can be changed at will. ***I-Scan*** supports a variety of approaches to calibrate sensors. The calibration used to make one movie can be stored independently of the movie, and used to calibrate other movies. In addition, a movie can be made only with raw Digital Output values. Then the sensor can be calibrated, and this calibration can then be applied to the existing movie. Some users will calibrate before making a series of measurements, and then repeat the calibration afterwards to confirm whether or not the sensor has drifted. If drift has occurred, the calibration values of affected movies can be edited.

Application Parameters to Mimic for Calibration

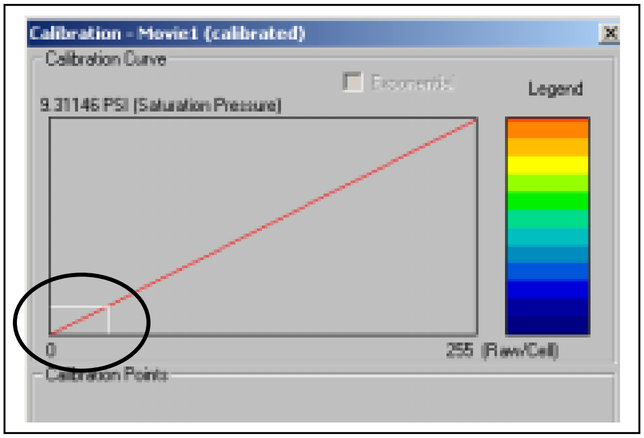
Pressure

Calibrate at a pressure near the working pressure or pressure of interest in the application. For example, it is not recommended to calibrate at 100 psi when you are interested in data at 10 psi.

Single-load calibration provides good accuracy for a range of pressures approximately +/- 20% of the single calibration load.

The white line(s) in the calibration dialog window graph represent(s) the average raw output(s) of the calibration point(s).

*Average **DO** of sensels loaded in calibration*



Material Interface

During calibration, the material interface should be similar to the interface of the application. For example, having the same durometer rubber in calibration and in the experiment is very important as the sensor has a topography that is sensitive to the effect of different durometers.

Interface Profile

If at all possible during calibration, try to mimic the shape or profile of the interface and impart similar pressure as in the experiment. For example, if the experimental interface is roller to roller, load two rollers onto the sensor pad during calibration.

Temperature

Ideally, keep the temperature during calibration similar to that of the application.

To illustrate the effect of temperature changes, experiments have been performed with a large temperature change, showing rather consistent output from the sensor. For example, a weight was put on a rubber pad bearing on a sensor in a thermal shock chamber. The temperature was dropped from room ambient temperature (72° F.) to -40° F and raised to 125° F. Meanwhile, the reported weight varied from 56 initially, and then 69 at low temperature to 56 at high temperature. Still, the temperature effect specification is broad and should be respected.

As a general guideline, for every ^{o}F. the calibration temperature differs from the application, the sensor reading can be off by up to 0.25%. In an automotive seat application, where a person's body temperature can affect the sensor output, calibrate at a temperature close to 98.6 ^{o}F.

Timing

To minimize the effects of drift, the calibration should mimic the time of the application (within reason). That is, if the measurement will be for 30 minutes of recording, you may wish to apply the calibration load for 2 minutes or more before calibrating. The system's drift curve levels off after approximately 30 seconds and then the 3% per log time effect results in small changes with increasing time. One can record a loading in raw digital output and then open a Force Vs. Time graph to view the curve to see where it begins to level off. For long duration static loads, the time after which the curve levels off (stabilizes) is a good indication of the time interval needed for calibration.

GENERAL CONSIDERATIONS

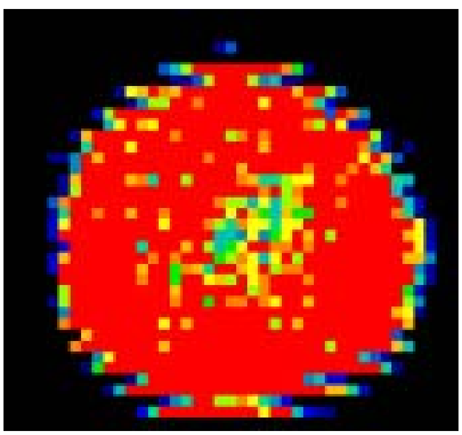
Calibration Load

Be sure the Physical load in the Calibration procedure mimics the application. Calibration pressure should be similar to application pressure. Preferably, load a large portion of the matrix area during calibration, and if this is not possible, load at least 25% of the active area to achieve statistically representative data. When small, hard parts are being studied, only a small portion of the sensor may be loaded.

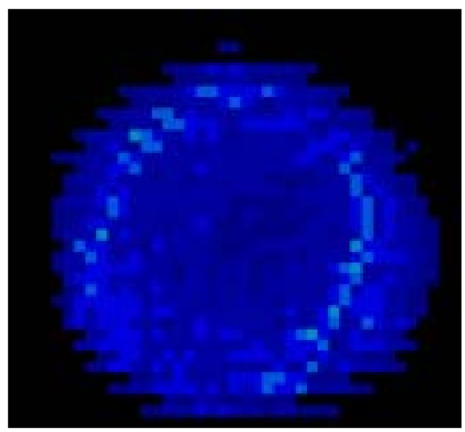
Ensure the load is balanced and stable before calibrating or testing.

Ensure when the sensor is loaded, and the legend set for a Raw Digital Output of 0-255, that the sensor is not reporting mostly red colors nor mostly blue colors. This would indicate the sensor is either too sensitive (red) or not sensitive enough (blue). If mostly blue colors or orange / red colors are displayed with the pressure legend set to full scale (0 to 255), then you may need to select a different sensor with a more suitable full scale range. The sensor's range should be such that at full load the display reports greens, yellows, and oranges; or 3/4 of full scale, allowing overhead for any pressure peaks during the experiment.

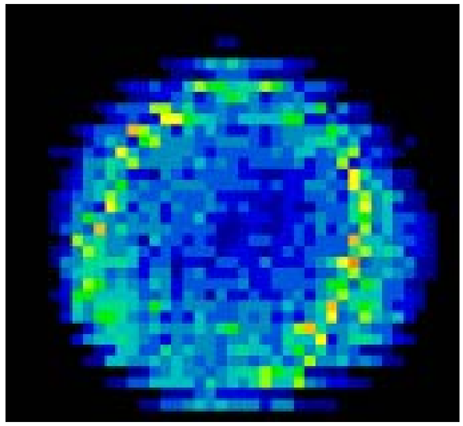
High Saturation (too sensitive)



Low Saturation (not sensitive enough)



Mid-Range Saturation (even color distribution)



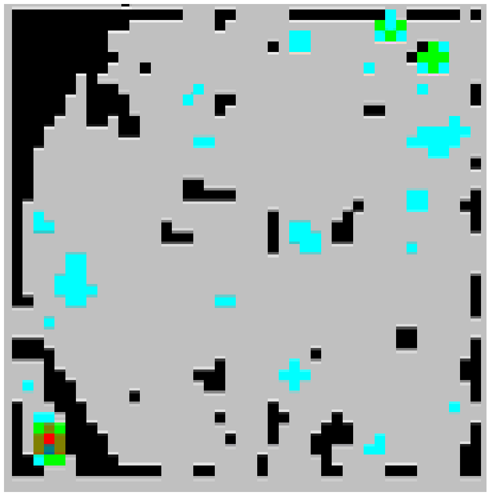
Sometimes an equilibration bladder is used both to equilibrate and calibrate a sensor. If calibrating with an equilibration bladder, the applied load is calculated by multiplying the applied pressure by the area of the sensor in which applied pressure is situated (as shown on the bottom of the real time sensor window).

When applying a very uniform load that is larger than the active area, be on the lookout for trapped air inside the sensor, which can produce a "pillow effect" (in which the trapped air bears some load) and not allow the two inner surfaces of the sensor to contact each other. In most situations, the surfaces to be used in experiments (the loaded area) are smaller than the sensor; so trapped air is not likely to be a factor because the trapped air is sent outward to the sensor edges where there is no load.

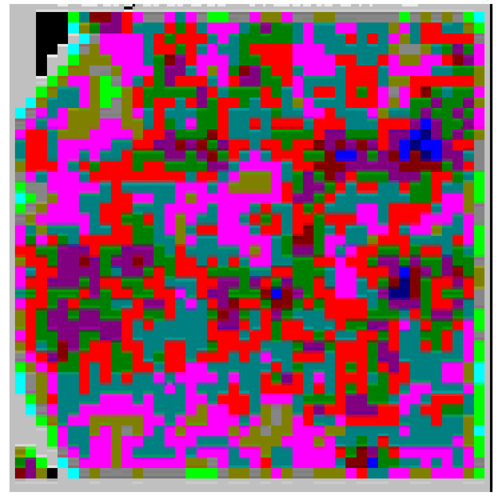
When trapped air is suspected, it usually leaks out over time. Many sensors have external vents to promote the release of trapped air. If the trapped air does not release in a timely fashion, pierce the substrate with a pin or a sharp knife. The hole should be away from any active traces or sensels. Many sensors have an internal vent, or pocket extending from the active region toward the handle. Piercing this vent near the handle may expedite the release of any trapped air.

Note: Even if the material interface inside the equilibrator differs from the actual application, the equilibrator offers relative repeatability.

The two images below show the output from a single sensor that is loaded with uniform pressure with entrapped air and without, respectively.



PSat Sensor Loaded with 10 psi Uniform Pressure with Entrapped Air



PSat Sensor Loaded with 10 psi Uniform Pressure - No Entrapped Air

Shear Forces

Tekscan sensors are intended to measure forces normal (perpendicular) to their surfaces. In most applications, as long as the substrate is not moved by the shear, the sensors are not affected by shear. In other applications, especially if the shear produces substrate motion, shear should be reduced or eliminated before it reaches the sensor. In these applications, over time, shear forces reduce the life of a sensor and alter performance. In dynamic applications, shear will produce noise, which can be seen on-screen. This affects output. Static applications do not produce this noise, and output is unaffected.

There is no calibration technique to account for or quantify the magnitude of shear. In most ***I-Scan*** studies shear is an artifact, and when shear is expected, most users take steps to reduce or eliminate this effect.

Secured shim stock can reduce or eliminate shear. If a piece of Shim Stock (discussed below) is placed over the sensor, and secured around its perimeter, shear is absorbed by the shim and not transmitted to the sensor. Teflon paper is an example of a material that allows the load to slide sideways without inducing shear on the sensor.

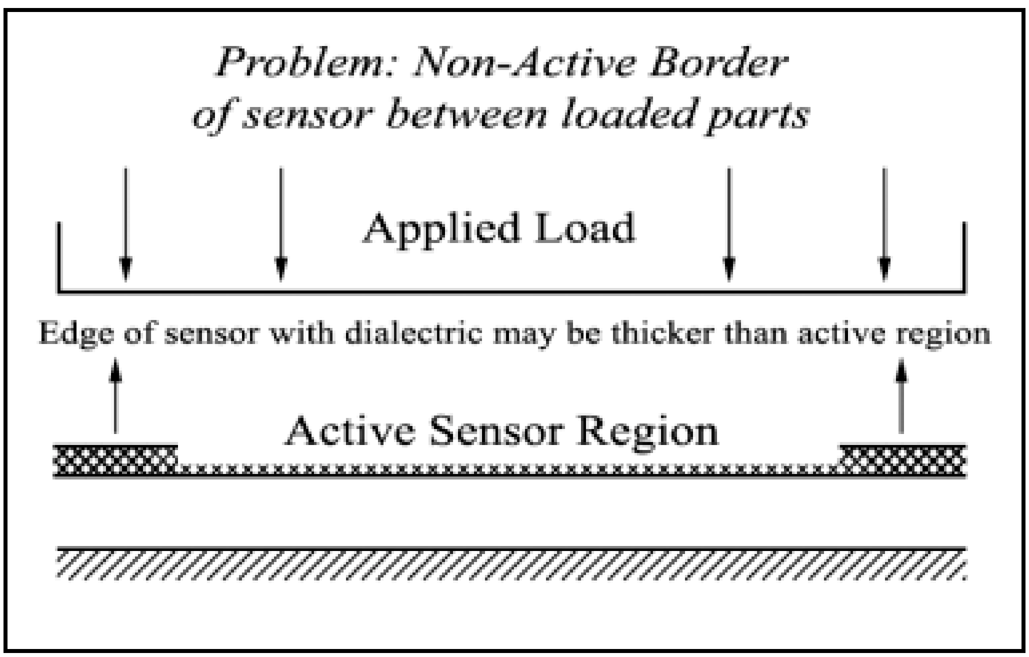
In other cases, the user may prefer to lubricate the surface of the sensor to reduce shear. Examples of lubricants include talcum powder, motor oil, silicon spray, etc. When working with tires and some rubber products, it is common to use soapy water as a lubricant.

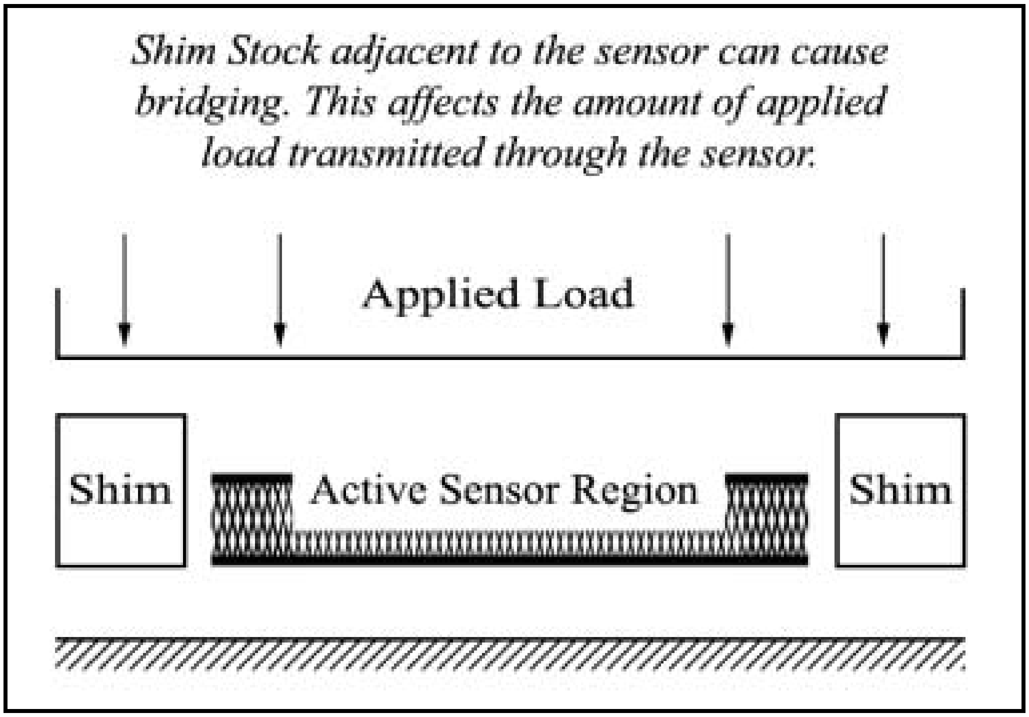
Shim Stock

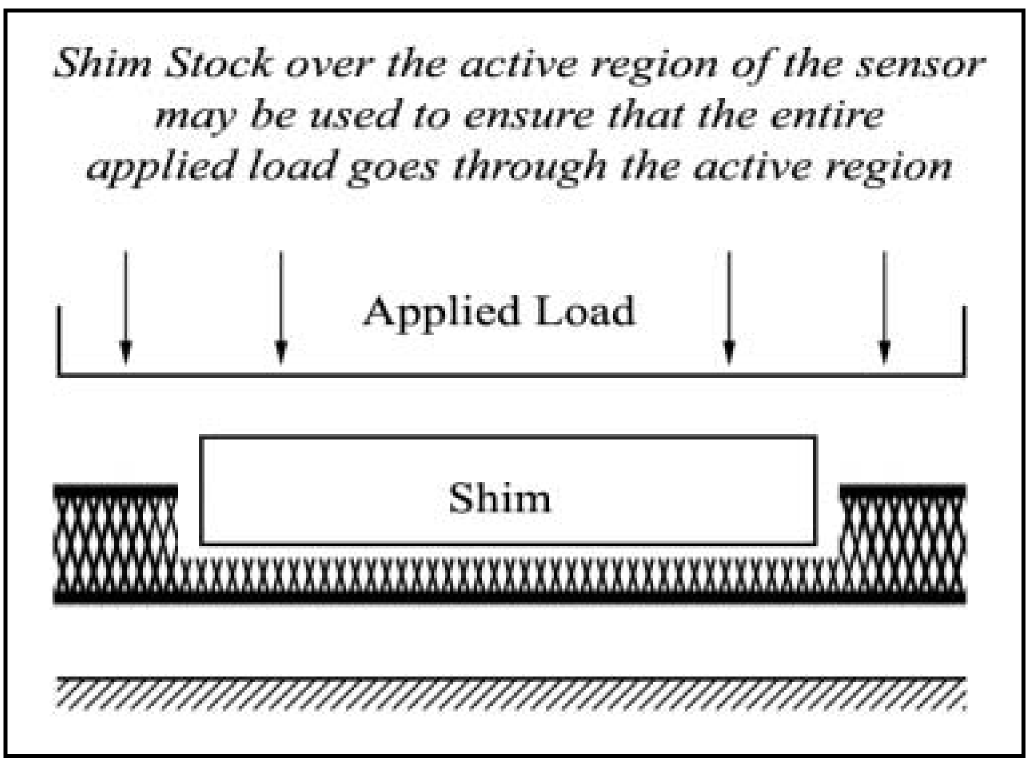
During calibration, make sure the entire calibration load passes through the active sensor region. Be careful no overhang causes the load to bear off the sensor. If there is shim stock adjacent to the sensor, please be sure that calibration force is not “bridged” by the shim stock *adjacent* to the active region of the sensor. Calibration force should instead *go through* the active region of the sensor. If the non-active border of the sensor (where there are connecting traces) is between the loaded parts, it is possible that some of the load is going through a non-responsive portion of the sensor.

If the application can tolerate insertion of thicker material, shim stock supports greater consistency of output when different surfaces bear on the sensor. To illustrate, consider the sensor between a soft and hard surface. The soft surface may “squirm” as load is increased, and this may result in an output increase from the sensor. By comparison, with the sensor sandwiched between two layers of shim stock, the squirm acts on the shim stock, and the sensor output is not influenced. With shim stock, the sensor consistently experiences loads through the shim stock, so the resulting measurements are more consistent, even when different materials are pressed together. A good rule to remember is if the experimental loads are coming through shim stock, the calibration should also be performed through shim stock.

Inserting soft shim stock between hard surfaces, such as metal plates, adds to the compliance between the surfaces. Pressure peaks are reduced and low-pressure areas are filled in. Hard shim stock, such as beryllium copper or stainless steel can preserve the original pressure distribution. In some cases, thin shim stock, such as a piece of paper may give significant protection to the sensor and still preserve the pressure distribution pattern. Shims may also be used if the hard interface overhangs the sensor. Shim stock, such as Teflon paper, photocopy paper or a thin piece of metal foil (stainless steel / beryllium copper), will offer more consistent sensor output. Calibrate with the same shim stock material as in the application.







Material Compliance

The classic definition of compliance comes from Hooke's Law: the relation between stress and strain. This defines the spring rate of a material. Put another way, for an applied force, how much does the material deflect? Materials that have large deflection from an applied force, such as foam rubber, are termed very compliant. Those that do not deflect much, such as steel, are termed non-compliant.

Most materials exhibit Poisson's Ratio behavior: the ratio of latitudinal to longitudinal strain, or the compressibility of a material perpendicular to the applied stress. Most materials squirm outward when a force presses down on them, maintaining relatively constant cross-section. For example, when you press down on a cube of Jell-O, the sides of the Jell-O cube are pushed outward the more you press down. In addition, if the top of the Jell-O cube is contoured, the contour will flatten along the top as you exert force downward. Most other materials exhibit this same behavior, more or less depending on the consistency of the materials, and the amount of force exerted on them.

Compliant materials tend to "cushion" high points of contact, and "fill in" low points of contact. They act as a spatial filter. In general, the effect of inserting compliant material (such as a rubber sheet) between two hard surfaces (such as two metal plates) is to lower the peak pressure areas, increase pressure in low pressure areas, and increase the contact area.

At a "micro level" there is a big difference between two hard materials coming together and a hard material bearing on a soft one, or soft on soft. Two soft materials give more contact area within the active area. Two hard materials give less contact area within the active area. When two hard surfaces touch, there often are three tiny regions with very high contact pressures, and large regions with no contact. The statement from high school geometry can be seen in action: Three points determine a plane.

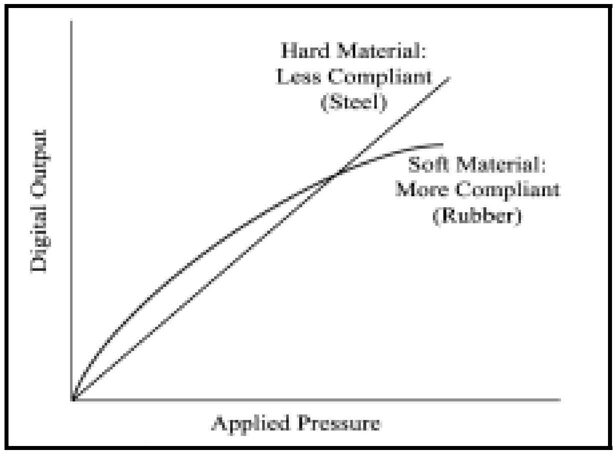
Imagine the deflection of a steel surface compared to the deflection of foam rubber with the same load. You can experiment with different materials and the same load to quantify the effect on contact area.

To elaborate on this issue, compare a bathroom scale to an array sensor for determining a person's weight. First, standing on a bathroom scale, the person weighs 200 pounds. The bathroom scale gives the same result whether the person is bare foot, in hiking boots, or hiking boots with gravel in the treads. The scale does not respond to pressure distribution.

Standing bare foot on a sensor array loads it with soft flesh that squirms outward with low pressure as weight is applied, increasing the contact area. There is moderate pressure under bones, but no high pressure points. With hiking boots, there is high pressure under the contact point of the lugs, and no contact pressure between the lugs. With gravel caught in the lugs, if the point of the gravel strikes a non-responsive area between sensels there might be no output. If the point of the gravel loads a sensel, it may saturate it. Saturated sensels contribute their limit to the total load, but no more, even when loaded two or ten times their limit.

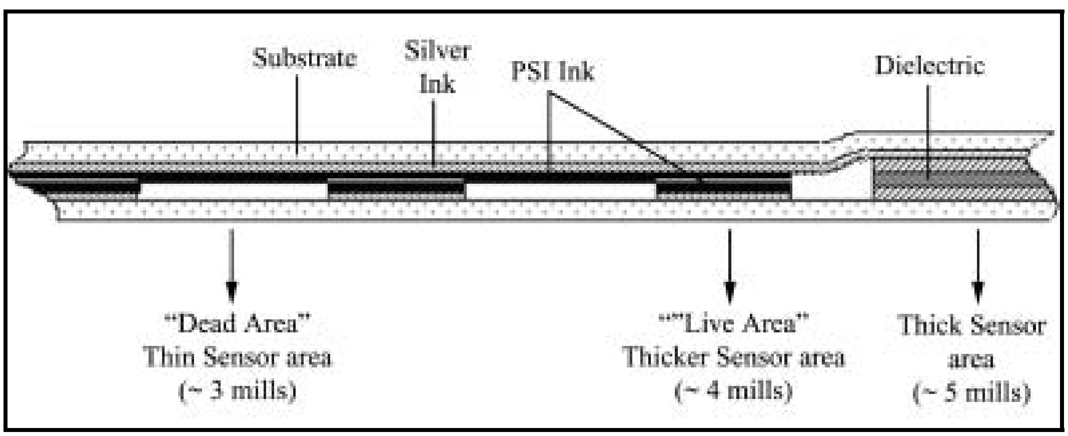
Another example of material compliance is placing clay on a window screen. The clay is soft, and therefore, very compliant. The points at which the screen grids meet and cross relates to the sensel grid, or sensing area. If you place the clay evenly on the screen with a low pressure, the clay will rest flat along the grid, and the crossing points support the load. The more pressure that is applied to the clay, the more the clay will seep through the screen (through the "empty" areas). Places where the clay extrudes through the screen relate to the non-sensing areas of the sensor. The empty areas are bearing some of the load. The relative change in output becomes smaller and smaller as more load is applied. The change in output diminishes with increasing load.

The following image shows the difference in output between hard (less compliant) and soft (more compliant) materials:



The image below shows a greatly exaggerated view of a hypothetical cross section of a sensor through both the sensing area and the non-sensing area. The left portion of the sensor is the sensing area and the thicker portion to the right is outside the sensing area. The black squares in the middle area are the sensels, while the white areas in the middle are non-sensing areas.

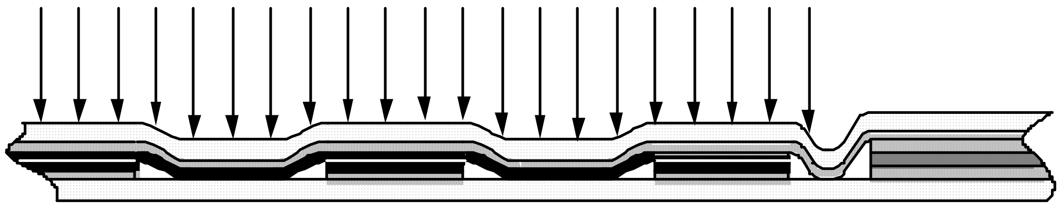
Edge View of Sensing Area:



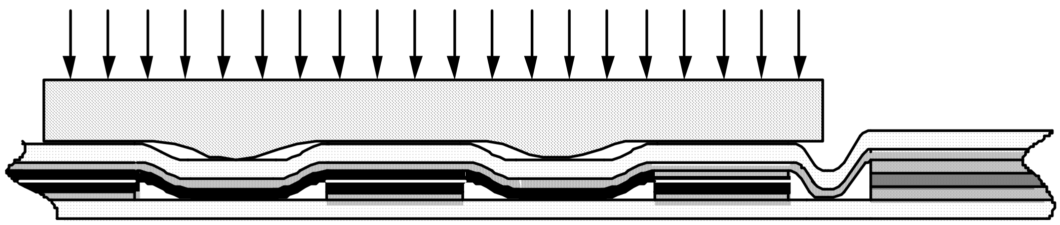
In general, for the section shown above, there are three basic stacking arrangements, in order of thickness. The thickest portion is the section that is outside the sensing area. The next thickest is the intersection points between rows and columns (where the sensels are located). The thinnest is the area between the intersection points in the sensing area (non-sensing area).

Note: some sensors' Dielectric Areas are thinner, which reduces the change in thickness along the sensor as a whole.

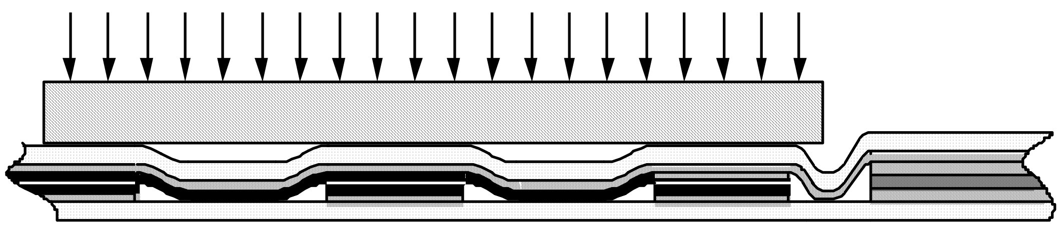
Sensor Loaded With Infinitely Compliant Hydrostatic Pressure:



Sensing Area Loaded With Moderately Compliant Material:



Sensor Loaded With Non-Compliant Material:



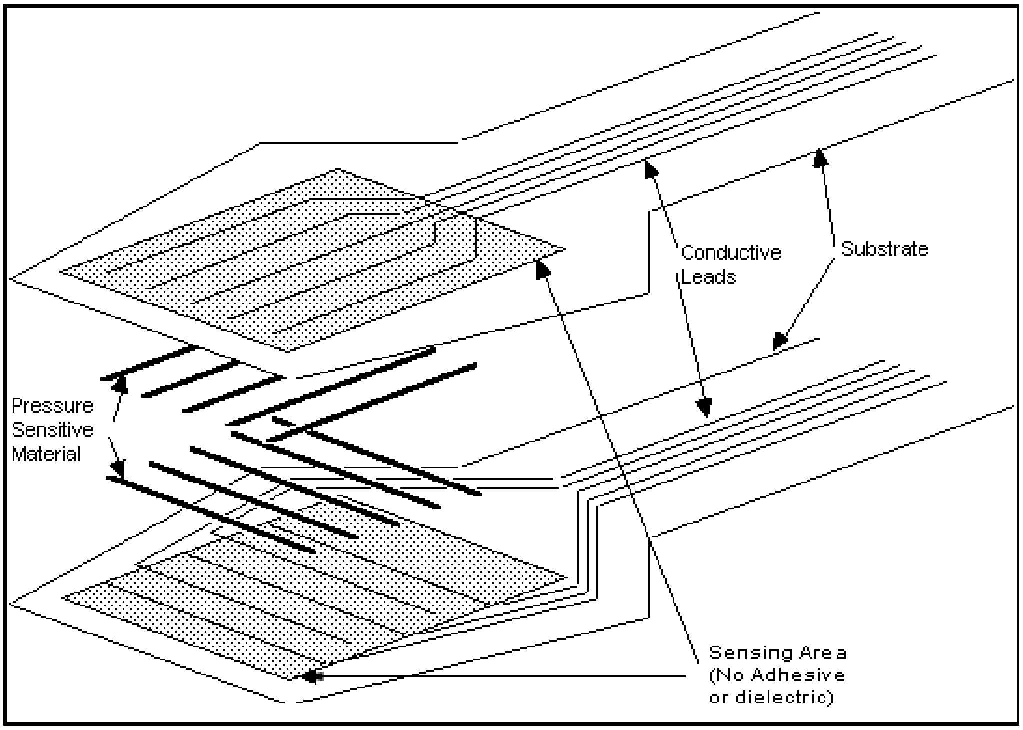
When loading the sensor, careful consideration needs to be given to the materials touching the sensor:

- When loading two hard, flat surfaces (e.g. two plates of steel), it is recommended to insert a sheet of paper, or thin urethane sheet between the two plates.

- When loading a part with high coefficient of friction and high squirm, such as rubber, onto the sensor, a thin sheet of Teflon paper works well to minimize shear forces, which can cause an output but cannot be quantified. Sensors are intended to measure compressive forces.

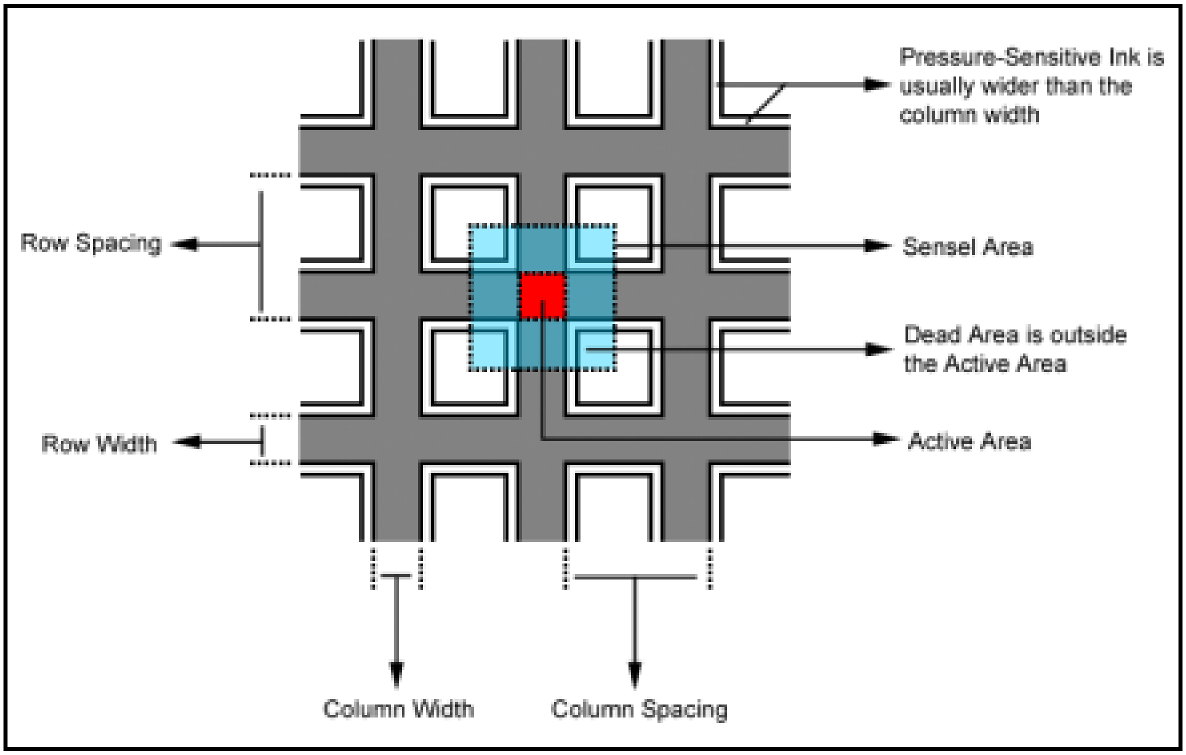
Active / Inactive Sensing Areas

Each sensel is a tiny load cell, whose resistance change is converted to force. The map, or software driver for the sensor, includes information about sensel area. Pressure is calculated as force / sensel area, displayed and reported on a sensel by sensel basis in arrays of data.



Within the sensel, a portion is “live”, and the balance of the area is “dead” or non-responsive. The live area of a sensel is where the silver trace of a column intersects the silver trace of a row. For polar coordinate sensors, the live area is the intersection of silver trace rings and spokes. The intersecting area is thicker, because, in addition to two substrate layers, it has four layers of ink: two of silver and two of pressure sensitive ink. The area outside of the intersection is non-responsive, because it lacks contact between two pressure sensitive layers backed up by conductive layers. The non-responsive area is thinner. Depending on which portion of the dead area is considered, it may have only the thickness of two layers of substrate, with no ink whatsoever. In addition to two substrate layers, it may have the thickness of one layer of silver and pressure sensitive ink. Other portions of the dead area may have two layers of pressure sensitive ink.

Consider the following diagram:



Each sensel has a measurable force, and the entire sensel area is used in the pressure calculation. Every sensel with a force above the display threshold contributes one unit of sensel area to the total contact area shown in the status bar.

The arrays of pressure data can be copied and pasted into other Windows applications, or groups of arrays can be exported in ASCII comma delimited form. The sensel area is included in header information presented above the array of pressure values. To convert the array of pressure data into an array of force data, multiply by the sensel area.

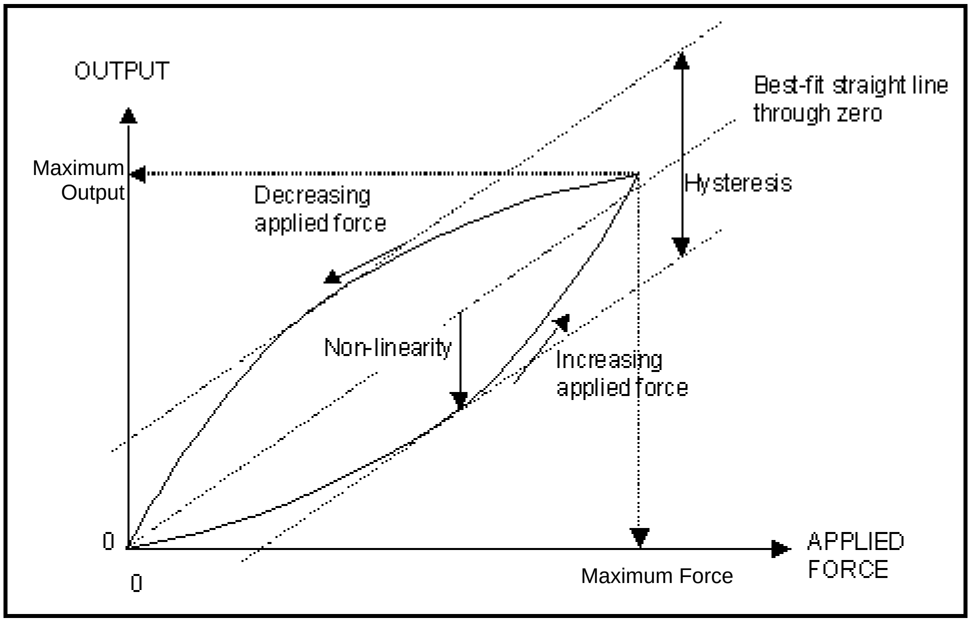
I-Scan includes various means of accessing and combining sensel data to report pressure. The user can point the computer cursor at a sensel and read the pressure of that sensel on the lower right main status bar. Clicking on Avg2 averages the value in each sensel with its eight neighbors, useful when pressure variations are expected to be larger than the sensel resolution. When an analysis object is placed on the sensor image, peak pressure for that image can be displayed. The factory default setting considers each two by two block of four sensels, determines the average pressure within the block, and reports the maximum value found within the analysis object as the peak pressure. Clicking on **Analysis – Properties** allows the Peak Area dX and dY to be increased, for example to 9 sensels in a 3 by 3 block. Because there is some variability in the response of each sensel to applied pressure, including more sensels in the calculation enlarges the area of the calculation and increases the confidence of the average pressure value.

Hysteresis

Hysteresis describes when the response of a physical system to an external influence depends not only on the present magnitude of that influence but also on the previous history of the system. Expressed mathematically, the response to the external influence is a double-valued function; one value applies when the influence is increasing, the other applies when the influence is decreasing. Hysteresis is not a repeatable phenomenon. It can fluctuate and produce different values each time, whether the load is increasing or decreasing.

Array sensors exhibit some hysteresis. The sensor is first loaded, and then unloaded, but does not follow the same path on its return to zero.

The following shows an example of a Hysteresis Loop Graph, as it relates to typical applications:



One way to overcome hysteresis is to mimic the application loading and unloading, as well as the duration of loading and unloading during calibration. The percentage of hysteresis in an application depends on the materials that come together. There is a contribution from the sensor of 0%-5%. In addition, the materials of the test, or the test apparatus is likely to exhibit hysteresis.

Duration of Load and Drift

Sensels respond rapidly to short duration or impact loads. At the other end of the time range, the sensor does exhibit drift with a constant applied load.

Drift is seen when a constant force is applied to an object. After a time, the reported force increases slightly. For example, think of a fixed weight placed on top of a spring. Over time, force acting on the spring is constant. Over time deflection increases slightly, as though more force were exerted on the spring. Thus the spring deflects downward a little further than when the weight was originally placed on the spring.

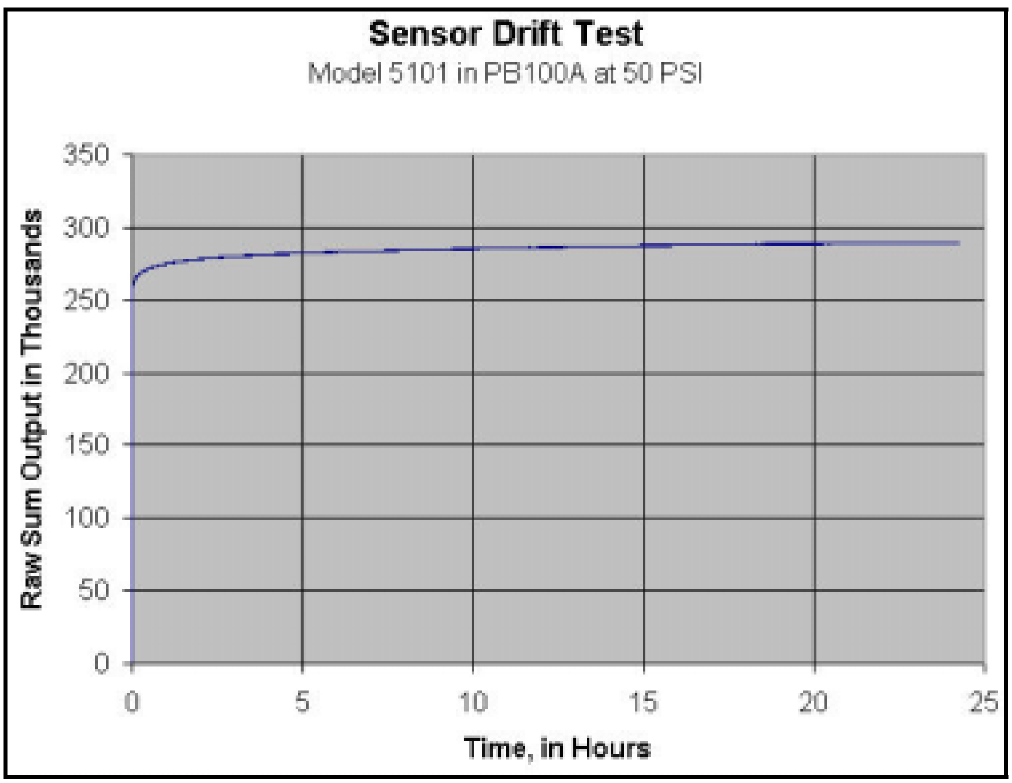
This phenomenon relates to our sensors in the same manner. The sensor exhibits drift with an applied constant load. The increase is an approximate range between 0%-3% per log time (in a worst-case scenario).

Example of Drift (Worst Case)

Duration of Load	*Output*
1 second	100 pounds
10 seconds	103 pounds
100 seconds	106 pounds
1,000 seconds	109 pounds
10,000 seconds	112 pounds

For example, if a 100-pound load is applied to the sensor; after 1 second the load would read approximately 100 pounds, after 10 seconds the load would read somewhere between 100 – 103 pounds, after 100 seconds the load would read somewhere between 100-106 pounds, and after 1000 seconds the load would read somewhere between 100-109 pounds.

Example of sensor drift:



Since most drift occurs at the beginning of the loading period, it is suggested to calibrate approximately 30-60 seconds after the load is applied or longer. When the system is shipped, the factory default setting is 30 seconds, and the user can change this in the calibration dialog window in the software.

Another way to reduce the effect of sensor drift is to mimic the duration of time of your application when you calibrate the sensor.

When doing a series of measurements, the trigger capability in the software can be used to achieve the same load duration, thus minimizing the effect of drift. For example, imagine two surfaces coming together in a press or load machine. ***I-Scan*** recording is triggered to start on first contact. The frame rate is set at one frame per second, and 60 frames are recorded. The user can view the “settling” of the two surfaces by looking at the first 59 frames. Frame 60 can be stored and recorded as representative of static conditions. Thus, the stored frame represents the same load duration from each test. Performing a few trials of this nature yields a series of frames where the frames are a consistent time after the trigger event.

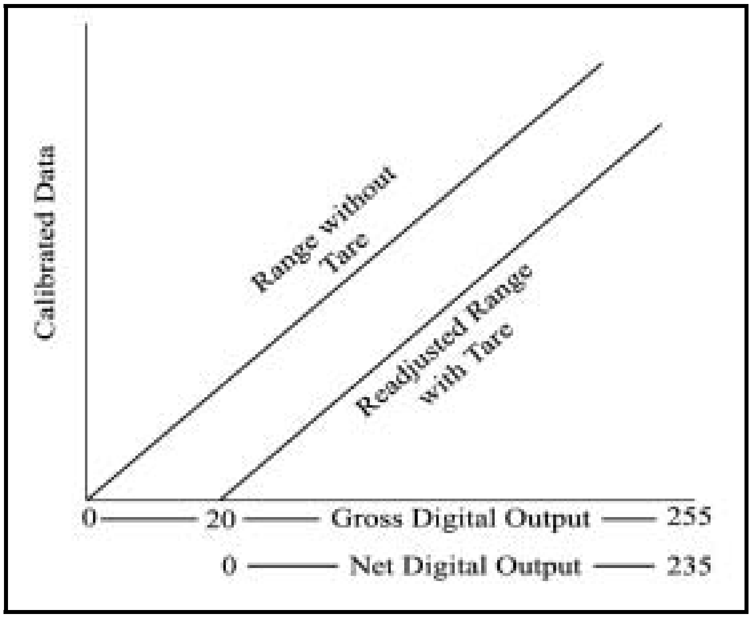
Tare

Tare is usually thought of as the weight of a carrier, container, or wrapper that is deducted from the gross weight of goods to determine the net weight of the contents. For example, you buy a container of goods from the supermarket. The total weight of the goods is 1.1 lb. The container weighs 0.1 lb. Tare is the action of removing the container weight from the total, or “gross weight”, in order to calculate the “net weight”. In this example, the gross weight (before Tare is invoked) is 1.1 lb., while the net weight (after Tare is invoked) is 1 lb. Put into a formula, it will look like this:

1.1 lb (before Tare) - 0.1 lb (container) = 1 lb (after Tare)
or
Gross Weight – Tare Weight = Net Weight

The way Tare relates to a sensor can be explained in the following example:

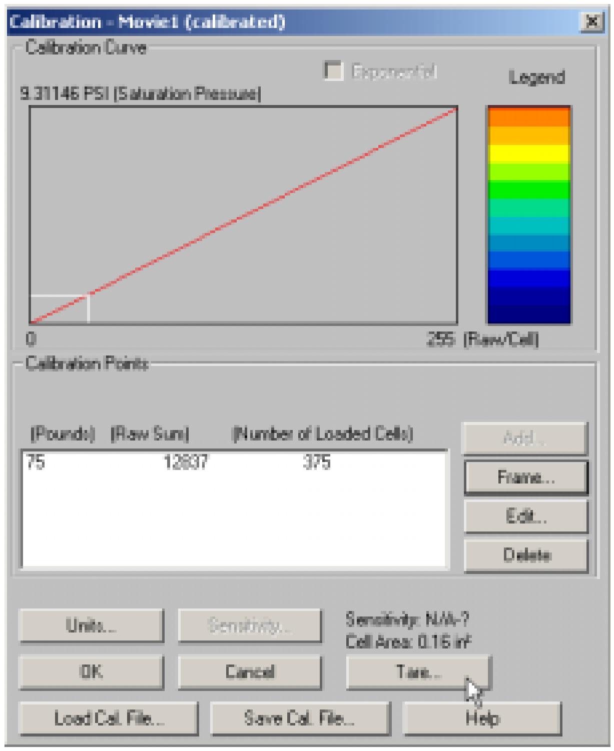
Sensors are flat by nature. Yet they are flexible, and so can be wrapped around a curved surface. When the sensor is placed on a curved surface (such as a grip sensor on the front of a hand), the sensor will read an output, though the output is a result of the curvature acting on the sensor, and not a result of any load applied (there is no physical load on the sensor). The Tare feature in the Tekscan software can remove this unwanted load and provide a “Net Force” as opposed to the “Gross Force”. When Tare is invoked, it operates on a sensel-by-sensel basis, and removes the force by raising the threshold from which Digital Output (DO) is converted to engineering units. The sensor’s DO range is 0-255. If one sensel reads 20 from unwanted influences, then 20 is subtracted from the range of the sensor, and that sensel’s net range will be 0-235 (255-20 = 235 pressure levels). Keep in mind that taring a sensor REDUCES the dynamic range that can be perceived by the sensor on the low end. Pressure level 20 is recalculated on the graph and becomes 0 instead, and 255 becomes 235 hence 0-235 (the only pressure levels that are removed are levels between 0 and 20 on the low end).

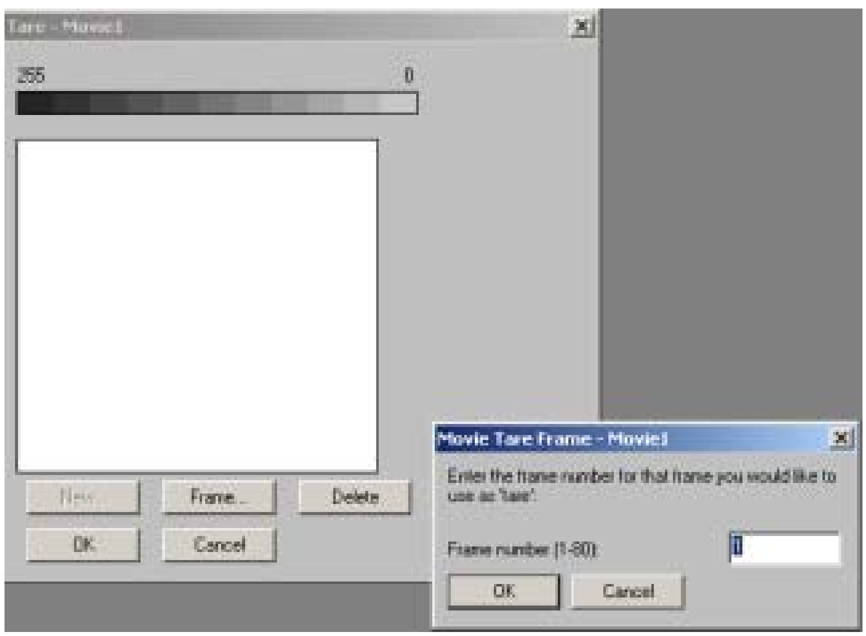


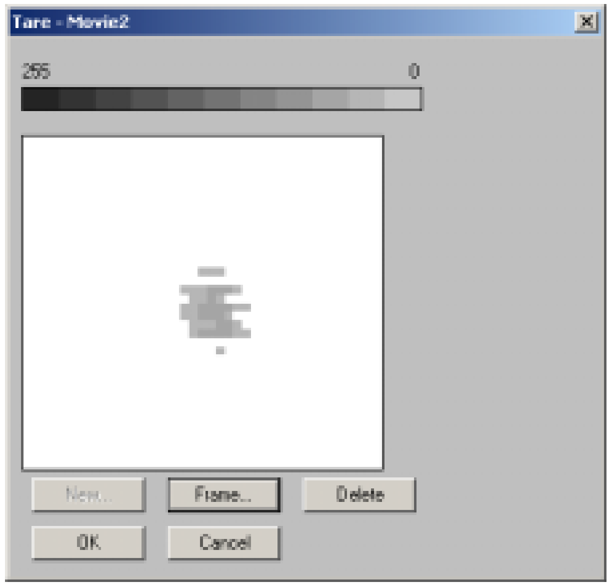
Consider another example. If a sensor with saturation pressure under 50 PSI is wrapped around a cylinder, the wrapping will induce some load, because the inner and outer skins of the sensor will rub against each other. After calibrating, go into the software and click on **Tools → Calibration**. Then click **Tare**. ***I-Scan*** allows a consistent output that is to be ignored (to be "tared out").

The amount of output due to wrapping depends on the full-scale range of the sensor. Sensitive ranges, such as 0 – 5 or 0 – 10 PSI give much more DO due to wrapping. Large ranges, such as 0 – 1000 PSI give much lower DO due to wrapping. Similarly, small diameter wrapping produces larger DO effects than large diameter wrapping.

Note: I-Scan can be toggled back and forth between gross weight and net weight by adding or removing the check mark in Tools → Calibration → Tare.


Calibration - Movie1 (calibrated)
Calibration Curve
Exponential
Legend
9.31146 PSI (Saturation Pressure)
0
255 (Raw/Cell)
Calibration Points
(Pounds) (Raw Sum) (Number of Loaded Cells)
75 12837 375
Add...
Frame...
Edit...
Delete
Units...
Sensitivity...
Sensitivity: N/A-?
Cell Area: 0.16 in²
OK
Cancel
Tare...
Load Cal. File...
Save Cal. File...
Help


Tare - Movie1
255
0
New...
Frame...
Delete
OK
Cancel
Movie Tare Frame - Movie1
Enter the frame number for that frame you would like to use as 'tare'.
Frame number (1-80):
OK
Cancel



There is an interaction between Tare and calibrated values. When working with net values, some DO amount is deducted from the gross DO value, then the DO is converted to engineering units. Since most surfaces coming together result in a non-linear conversion from DO to engineering units, the net values typically lose accuracy.

Note the following:

- Each sensel is tared individually.
- There must be a minimum of 3% of the sensor area loaded in order to invoke Tare.
- Tare only works on calibrated data. That is, your movie must be calibrated before you can utilize the Tare feature.
- Tare will only be seen on calibrated data (ie: engineering units of measurement). Tare cannot be seen when looking at the Raw Digital Output data.
- Large tare values used with a non-linear calibration will cause errors in reported engineering units. Because single-load calibration provides a straight line, the effect is much less severe with single-load calibration.

APPLICATION CONSIDERATIONS

Low Pressure Application

With low equilibration pressure, there is low drive pressure to force trapped air out of the sensor. So when a low equilibration pressure is applied to the entire active region of a sensor, the air trapped inside the sensor may form a "pillow." The resulting screen image will be very non-uniform. Subsequently, when the sensor is removed from the bladder, outside air pressure acts on it, and if the internal air has been squeezed out, sensels may register pressure when only atmospheric pressure is acting on the sensor.

For example, imagine applying new wallpaper to a room. Bubbles of air become trapped between the wallpaper and the wall. The back of the wallpaper does not touch the wall. In the same way, sensels with trapped air underneath will either register no pressure, or low pressure.

1. One approach is to take multiple readings to reduce error. Perform the same test multiple times, and then average the results of the test. This will provide a more accurate result.

2. The second approach to this is to wait. Let the trapped air escape when the sensor is inserted into the bladder, and wait for air to re-enter the sensor after it is removed from the bladder.

3. The third approach is to insert only a portion of the sensor into the bladder. Then the trapped air can move to the region outside the bladder.

4. As an alternate fourth approach, with a pin or sharp knife, make a small hole or holes in between traces of the sensor to allow trapped air to vent or escape and later to allow outside air to re-enter.

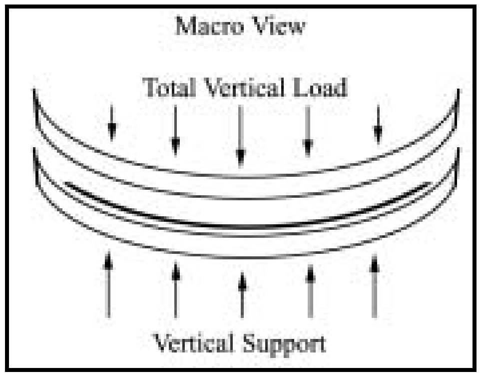
Calibration at High Pressures

It is challenging to achieve uniform pressure distribution with hard surfaces loaded to high pressures. If the calibration involves hard materials and pressures above 100 PSI, are you using a load machine for calibration? Many users have an arbor press with a load cell, Instron, MTS, Tinious Olsen, or similar load device. When used in compression, it is important that the opposing plates that come together are parallel; otherwise one, two, or three individual points will have very high pressure, and most of the apparent contact region will have no pressure or very light pressure. Some load machines have a concave dish and matching convex load plate to allow adjustments to obtain a parallel situation. If you are using one of these, please ensure that the dish is not binding and is properly lubricated. Also, many load plates have central holes for attaching tension devices. In compression, these holes often result in regions without any pressure. Stiff metal blocks can possibly bridge the holes. Even if the surfaces that come together are relatively hard, it is suggested that compliant material be used in another portion of the stack to provide compliance or a spring effect to distribute the load as uniformly as possible. For example, a urethane sheet could be used.

With pressures above 10,000 PSI, the pressure sensitive ink used in the sensor becomes plastic, and in the worst-case, “creeping” or “drifting” occurs. If the test load repeatedly hits one portion of the sensor, that part becomes less sensitive, and as a result, the sensor becomes uneven. One approach for high-pressure measurements is to precondition the sensor with a carefully applied, uniform high pressure. The initial drift then occurs before the experiment begins. Another approach is to match the time the load is applied during calibration to the time the load is applied in the experiment.

Curved Contact Surfaces

When studying contact between soft or curved objects such as the foam of a bed or chair and human flesh, or artificial knee joints, calibrate with the objects being studied, not with an equilibrator. That way the curvature that evolves between the surfaces in calibration is similar to the curvature of the experiments. To better understand this, consider a free body diagram of a concave curved surface supporting a convex vertical load. The total resulting vertical force vector results from many individual local vectors with opposing horizontal components, which cancel out. Tekscan sensors respond to the individual local normal forces, and the software sums these forces (assuming the sensor is flat).



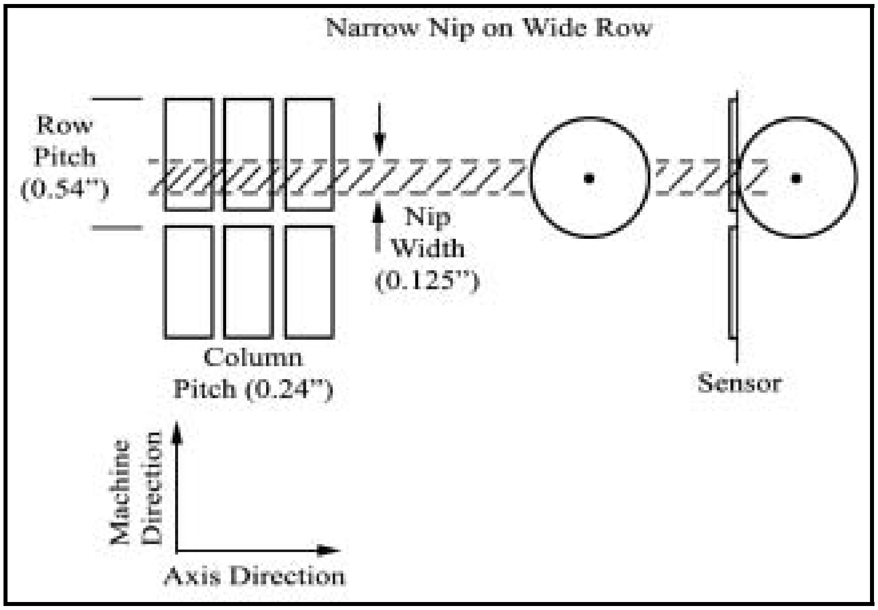
Tekscan sensors are made with dimensionally stable substrate: polyester sheet. A dimensionally stable sheet rolls nicely around a simple curve such as a cylinder or cone. Thus, measurement can easily be taken on a coffee cup, rolling pin, or the body of a ballpoint pen. A dimensionally stable material has difficulty wrapping smoothly over a ball, or someone's nose or chin. If the sensor crinkles, or develops folds, it is prone to reporting high-pressure output where little or no pressure is applied. These locations may be "tared out," the bottom threshold of the legend can be raised, or they can be edited out of the movies altogether.

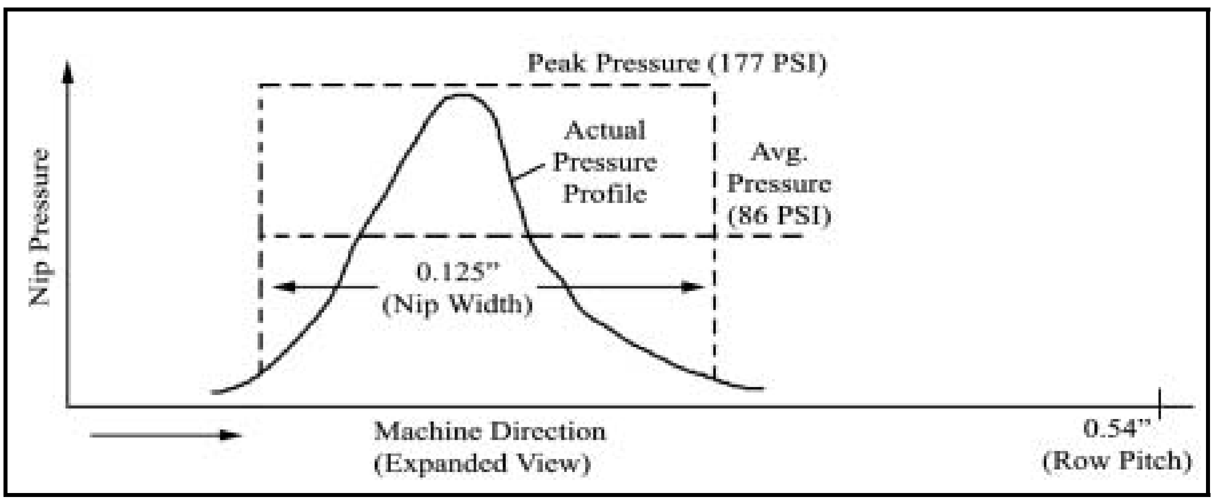
When loading soft interfaces, such as a crash dummy or a person onto a foam/cushion, try to calibrate with the actual materials and with the actual profiles/curvature. Curved contact surfaces with complex curves may be measured with Tekscan's sensor models 6900, 9801, or 9830 since they have narrow sensing areas ("fingers") that lend themselves to positioning on contoured surfaces with minimal artifact generated from the geometry.

Wide-Row Nip Measurements and Small Contact Areas on Big Sensels

Nip measurements with sensors such as 5501, 5560, 5570, or 5580 present a special calibration case. These sensors have wide rows, with the intent that the entire nip hits only one row, with plenty of spare row width to make alignment easy. Therefore, the applied contact area may be much smaller than the row width. Equilibration is especially important, because each column along the axis of the nip has only one active row. Since the software calculates output based on the active area of the sensel, and the area of nip contact is smaller than the sensel's active area, the actual pressure of the nip may differ from the software's calculated output.

Example of a narrow nip on wide row:





To expand on this point, consider sensor 5580 used on a nip that is 1/8" wide in the machine direction. The row spacing is 0.54" and column spacing is 0.24", so sensel area is 0.1296 square inches. If the sensel reads 20 PSI, the total sensel load is (20 lb/in^2) * (0.1296 in^2) = 2.592 lb. This force acts over 0.24" along the axis, so there is 10.8 PSI. However, the force is actually applied over 0.125" in the machine direction, so the average pressure is (2.592 lb) / (0.125" X 0.24") = 86 PSI. Since most nips have a "bell curve" pressure distribution in the machine direction, the peak pressure is about twice the average pressure, or about 177 PSI.

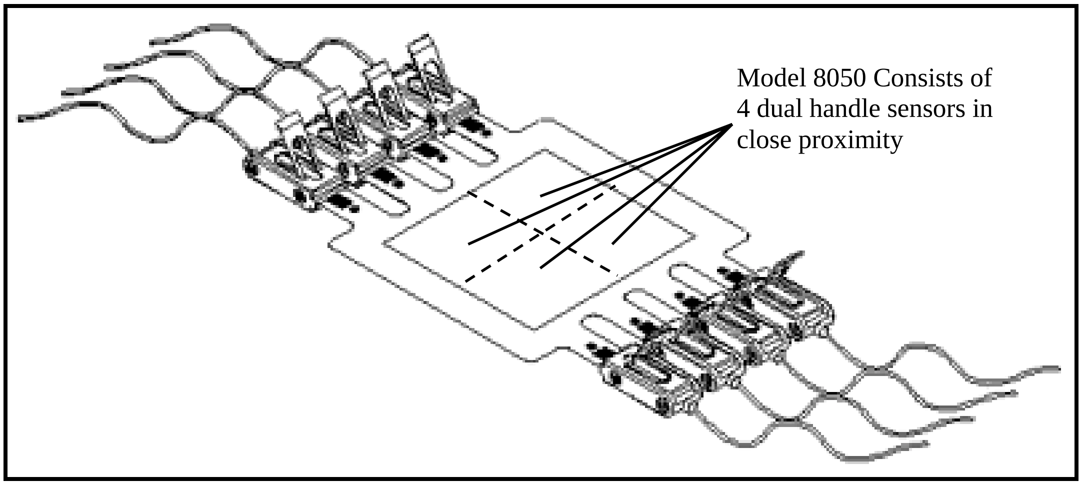
Use of sensors with high-resolution rows, such as the 5526 may be useful for an alternate study of small nips. With the 5526 sensor, care must be taken to ensure proper alignment between the rows and the nip. For nips with a width that can cover 5 or more rows, row resolution such as sensor 5526, 5555, or 6300 are preferred.

The same effect operates whenever the applied load is smaller than the active area of a sensel. For example, imagine pressing the tip of a ballpoint pen on a 5315 sensor with a force of 2 pounds. The tip of a pen may have a contact area 0.020” in diameter (0.5mm). So the average tip pressure is 6,000 PSI. The 5315 sensor has rows and columns every 0.4”, or put another way, a sensel area of 0.4 x 0.4 = 0.16 square inches. It would report a pressure of 5 PSI (its saturation pressure) when the pen is on the active region of the sensel, and 0 when the pen is not pressing where the silver traces cross (in the “dead” area).

Multi-Tile Calibration

If a multi-handle system (e.g. four handles, each with its own sensor, such as four 5051 pads) is being used, one may calibrate each sensor individually or calibrate all four simultaneously. If a four pad sensor is being used (e.g. 6900 model), one may calibrate each sensor pad individually as a calibration region or calibrate all four simultaneously. This feature was introduced in ***I-Scan*** version 5.10.

Sensor maps that include calibration regions have the suffix “CR”.



Wet Environments

For limited times, most sensors can be exposed to water or liquids, such as motor oil. The substrate of most sensors is a 1 millimeter thick polyester sheet, which is a good barrier to moisture and most liquids. The top and bottom substrate at the edge of the sensor is joined together with an adhesive. Eventually, water or liquids will enter the sensor, either through the adhesive or through the substrate. Some sensors have vents that allow air trapped inside the sensor to escape when a high percentage of the active region is loaded. Water or liquids will quickly enter through these vents. Since the top and bottom of the interior of the sensor must have high impedance between them to function, entry of even a tiny amount of water or liquid will end the useful life of the sensor.

If the sensor is to be exposed to water or a liquid for an extended period of time, use of a barrier material or shim stock is recommended. For example, the sensor can be inserted into a plastic bag, and the bag can be sealed against liquid. Thus, the sensor measures contact forces in a liquid environment, but is kept dry.

A protective enclosure, or “elephant capsule” may be used to protect the handle from water or drop damage. The “elephant capsule” is a plastic capsule with an internal web that should be cut down with a dremel tool or similar device. Cutting the web allows the sensor tab to connect to the handle. Then the perimeter of the capsule can be sealed with caulk or RTV.

Right Angles and Wrinkling / Bending

If a sensor is pierced, cut, severely sheared, wrinkled, creased, or repeatedly bent in the same location, the sensor may exhibit an electrical open and / or electrical short, or may simply report artifacts instead of pressure. Always attempt to minimize sensor wrinkling and sharp bending as this will prematurely wear the sensor.

For right-angle applications, use two sensors, if possible. If a single sensor is wrapped around a sharp edge, artifact will result at the bend line. The stress of making a sharp bend may fracture some silver traces.

Applications Using a Vacuum

I-Scan interprets resistance levels greater than 10 Mega Ohm as having no load. If the vacuum causes the sensor substrate to separate, its impedance may rise above 10 Mega Ohm. The system detects this as “no load,” and does not respond with an output. If the sensor has air trapped within it, the substrate may be pierced with a pin or sharp knife to allow a path for the air to escape.

Impact Load Calibration

Impact loads present a particular calibration challenge. It is very difficult to arrange a known calibration impact force with the same duration as a rapid event. If an object of known mass and incident velocity strikes the sensor with known dwell time and known exit speed, Newtonian physics can be used to calculate the peak force.

Impacts may affect the sensor as briefly as a few milliseconds. In calibration, applying a dead weight load to the sensor takes seconds to minutes, so that method does not mimic the elapsed time of the experiment. Some researchers place high-speed load cells in series with a Tekscan sensor to measure events of a short duration. When that is not convenient, consider placing the sensor on a plate at the bottom of the swing of a pendulum. If the mass, impact velocity, and rebound velocity of the pendulum is known, force can be estimated from Newtonian mechanics (Force = Mass *

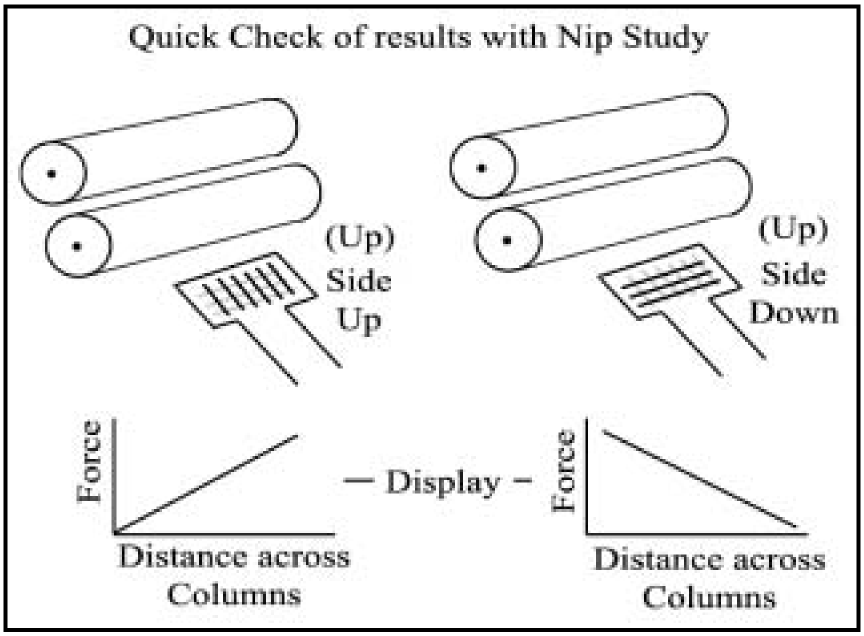
Acceleration). Other researchers mount an accelerometer in the pendulum to obtain a direct reading of acceleration.

Impact loads (High Speed) may be calibrated after the recording with Frame (Post) Calibration.

Quick Check of Results

If the pressure pattern on the computer display shows uneven loading, and you question whether this is actually representative of contact between the parts, or due to an uneven output of the sensor, here is a quick check:

Remove the sensor from the contact region, invert it, and re-insert it. Compare the original image with the one from the inverted sensor. If the uneven pattern stays the same on the computer screen, the uneven output is most likely due to the sensor. If the uneven pattern reverses, it is due to a physical contact issue. This is an especially relevant check when doing small nip studies. Similarly, if the loaded region is small, repeat the experiment with the load on another part of the sensor.



If there is a flip in display when the sensor is flipped, then the sensor is ok, and the problem could be mechanical.

SENSOR CONSIDERATIONS

Determining Sensor Range

Tekscan uses two different techniques to measure sensor range, depending on the full-scale pressure. For sensors to be marked 125 PSI and lower, a portion of the sensor is inserted into an equilibration fixture that has an internal urethane bladder. Inserting a portion of the sensor into the fixture allows trapped air to move to the part of the sensor that does not have applied pressure.

Sensors to be marked over 125 PSI are tested in a load machine that applies force through a puck that has a urethane core capped with a steel surface that bears on the sensor. The urethane core is compliant, providing relatively uniform applied pressure. If the puck were solid steel, it would be likely that two or three individual points would have very high contact pressure, and the rest of the area under the puck would be un-loaded. Due to the compliance, minor mis-alignment between the plate and plunger of the load machine does not result in extremely large pressure on one side of the puck, and no pressure on the other. Small sensors are loaded with a puck with one square inch of area. Larger sensors are loaded with a four square inch puck. An analysis box is drawn around the loaded region, excluding partially loaded sensels at the edge of the puck. The average raw Digital Output (DO) value inside the analysis box is used in the determination of range.

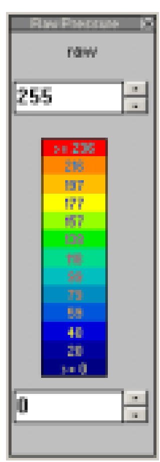
The sensor is loaded with five sequential pressure values, relatively evenly spaced, up to a DO of about 180. The data is fit to a power law curve, and checked for expected linearity. By extrapolation, the pressure associated with a DO value of 200 is labeled as the full-scale range. This technique allows some “head room” for tests that have locally higher pressures.

Sensor Saturation

The pressure range of the sensor should be large enough that the sensor can register all (or at least most) of the distributed pressure values in the experiment. The sensors should not saturate with a full load, or at the very least, the saturation of sensels should be minimal. The finest resolution is achieved by having a sensor with sufficient range to capture the peak pressure without saturating. This makes sensor range selection important with a fixed gain system. With an adjustable gain system, run the experiment with a trial gain, then adjust the sensitivity up or down to optimize sensor behavior.

To check for sensor saturation, observe the *Raw Digital Output (DO)* from the sensor. Go to **Options → Measurement Units** and select **Raw** for the “Units of Pressure.” Since Tekscan uses an eight-bit “Analog to Digital” Converter, the maximum output of any sensel is a Raw DO of 255.

Ensure the pressure legend is Open and set to full scale Raw DO (0 to 255) before calibrating or simply for testing to see if the sensor PSat range is suitable.

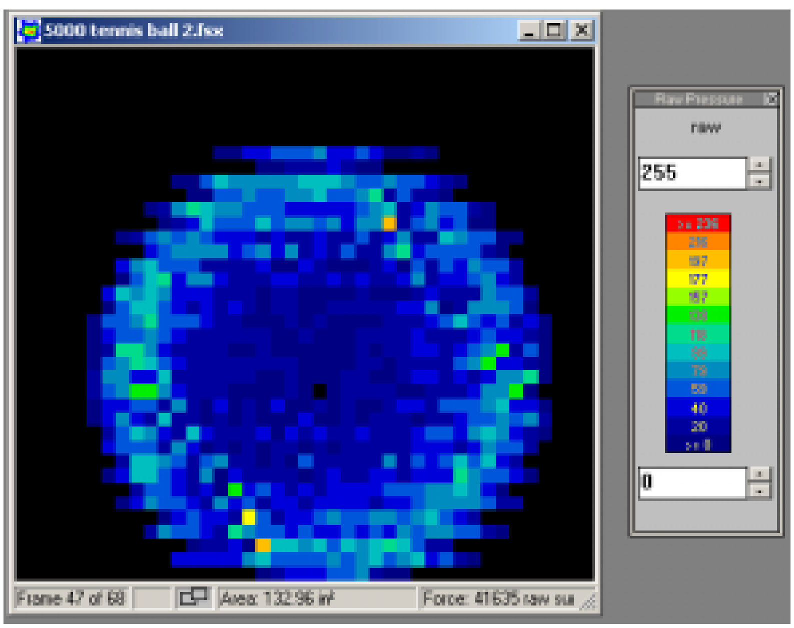


For optimal resolution, the peak load during the experiment should nearly saturate the sensor output. In a typical experiment, the maximum DO should be approximately 220, which allows for "headroom" for any circumstances that yield a higher force. A sensor that yields a display of mostly red or all red most likely has saturated sensels, therefore, the data is suspect.

After a sensel reaches saturation (a Raw DO of 255), the load on it can double, or go up by a factor of ten; however, the sensels will not be able to capture any higher load. If the sensor exhibits saturation of sensels, then a sensor with a larger range, or reducing the sensitivity of the sensor (with an adjustable gain system) is suggested.

Looking at the screen image of the example load, put the cursor on any red areas of the sensor image. On the status bar in the lower right corner of the screen you will see the raw Raw DO value of these red areas (this information changes according to the position of your cursor on-screen). If the status bar indicates raw Raw DO values above 240 – 250, a sensor with a higher saturation range should be considered.

*Example of a Tennis Ball on the **I-Scan** sensor, along with the pressure legend's Raw DO*



Note: The PSAT (Saturation Pressure) written on the sensor is a guideline, not an exact specification for every test since the material interface Tekscan tests with may differ from the experimental material interface.

When an adjustable gain system saturates, the sensitivity should be reduced. Alternately, you can change the bottom value of the legend to 240 or 250 (instead of reading 0). This will show only the sensels whose DO is greater than 250. This method provides an easy way for you to determine the number of sensels that are saturated, or close to being saturated.

As an example, if the pressure of interest is 75 PSI, the gain might be set to give a DO of 125 where the applied pressure is 75 PSI (or approximately halfway within the sensor's range). Then the maximum pressure that can be measured will be about (75 PSI) (255/125) = 153 PSI. This provides "headroom" for experiments with a significant pressure variation.

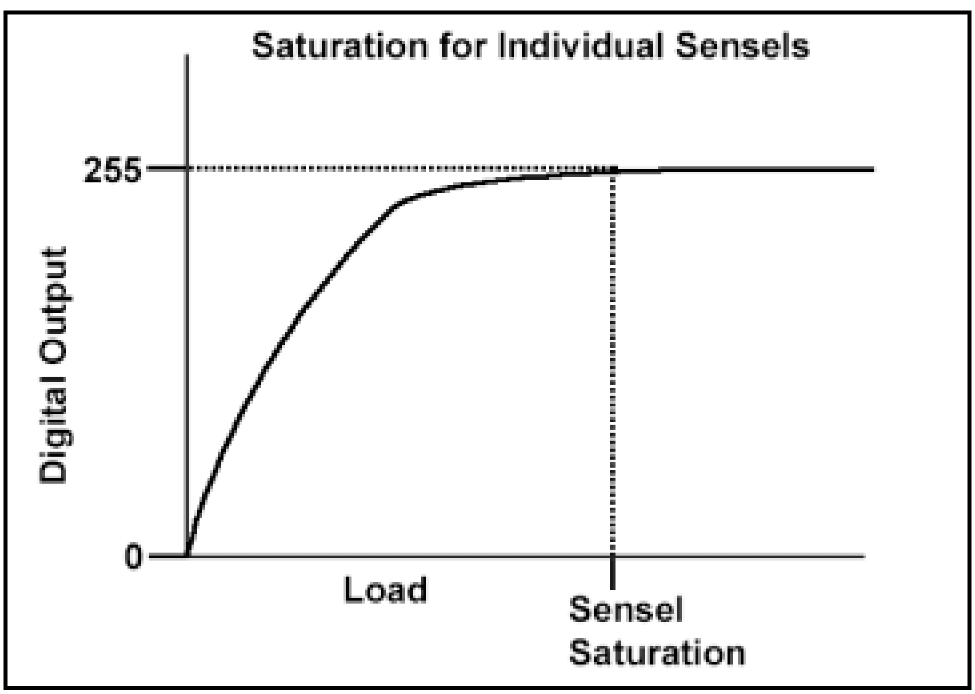
Conversely, if during calibration, the maximum DO from the sensor is below 50, the sensor has too much range. Increase the sensitivity with an adjustable gain system. If, during calibration, the maximum DO is 50, and the experiment involves DO values above 100, the engineering unit results may not be as accurate as the previous example where the pressure of interest is 75 PSI because they come from extrapolation beyond studied values.

Overloaded Sensels

Users often ask if overloading damages the sensor. As long as the substrate is not cut or badly dimpled, creased, or folded, high loads do not permanently damage the sensor.

Sensel saturation does undermine the theory of operation: that each sensel is independent of its neighbors. The "no-load" impedance of each sensel is 10 to 20 Meg-ohms. With no load, or light load, each element is independent of the other. However, if one or several row/column intersections are shorted, alternate current paths may be set up in the sensor. This could cause regions without load to appear to have load. This can cause the "bleed" of a pressure image from one portion of the sensor to another.

The following shows the relationship between the Sensels and Saturation. A Raw Digital output of 255 is the maximum Saturation for each individual Sensel.



Adhesives

In many experiments it is advantageous to secure the sensor in position. Sometimes it is sufficient for the perimeter of the sensor to be held with tape, such as cellophane tape, masking tape, or duct tape. When an adhesive is applied to the active region, please be aware that most water-based adhesives (such as white wood glue) will not stick to polyester sheet or Kapton. Spray adhesives, such as those used by artists, adhere well to the sensor.

One choice is 3M's Adhesive 77, available at most office supply companies. Spray evenly and lightly. Do not allow any of the adhesive to clump, or a pressure artifact may result.

Another adhesive to consider is double-sided tape. If tape is used to secure the sensor, be careful not to cover the active region of the sensor since a pressure artifact may be created where there is greater thickness as a result of the tape, or an overlap of two tape layers. When tape covers some, but not all of the active region, apparent low pressure areas may appear where the stack is thin.

Sensor Turn-On threshold

The software's factory default setting of the noise threshold is three digital raw counts (out of 255). If a significant amount of random noise is present, it is recommended to increase the lower limit (field) in the Pressure Color Legend, rather than increasing the noise threshold. Keeping the noise threshold low allows all possible data to be captured in the recording. Raising the bottom legend value causes lower values to be omitted from the display: the Pressure Color Legend serves as a visual filter. Alternately, The Tare function can remove a preload or pattern seen on the displayed image.

In addition to software threshold, sensors have a physical turn-on threshold. For most sensors, this is about four grams of force per sensel.

Representative Sensor Turn-On Pressures		
Full Range	*Pressure Measurement Resolution*	*Turn-On Pressure*
5 PSI	0.02 PSI	Depends upon sensel size
25 PSI	0.1 PSI	0.3 PSI
250 PSI	1 PSI	3 PSI
2,500 PSI	10 PSI	30 PSI

For example, measuring the pressure of applying cosmetic make-up to a face involves delicate forces. The force between the applicator and the face is too low for our sensors to measure accurately (it is below the sensor's turn-on threshold).

Stacking Sensors

In some cases, sensors are designed to be stacked one on top of another. For example, sensor 9550 has discrete sensels with large open spaces around them for a 25% active area. If four are stacked with careful offset, the sensels provide 100% coverage of a surface. When stacking model 9550, care must be taken to obtain precise alignment, and to deal with trapped air. Since each sensor has different surfaces bearing on it, each will have different output from what might appear to be the same load.

In most cases, however, stacking sensors one on top of the other for a measurement is not recommended. For example, a researcher places one sensor with a low Saturation Pressure (PSAT) on top of the same sensor model with a high PSAT in an attempt to obtain the best possible pressure measurement resolution. The intent was to get high resolution of low pressure areas from the sensor with low PSAT. Areas with high applied pressure were to be studied with the sensor with high PSAT. Unfortunately, the low PSAT sensor becomes highly saturated, degrading its performance. Also, since sensors do not have the same thickness everywhere, the output is strongly affected by slight changes in alignment. Alignment issues are described as a Moire effect.

Cutting the Sensor

Some sensors lend themselves to being cut; that is, some sensor designs allow for removing part of the sensor with an exacto knife, hole punch, or scissors resulting in no loss of data from the remaining sensor region after the cut. If you trim the sensor, protect the exposed inner portion of the sensor from any dirt or liquid. Usually a piece of tape works well.

The white stripes visible in the sensor are conductors, essentially wires routed out from the handle. Their distal end can be removed, and since no load is interpreted as a high impedance, the portion removed does not register any pressure. The cut edge should be re-sealed with tape to prevent dirt, water, or contamination from entering inside the sensor (as described above). The opposing sides of the sensor have a “no-load” resistance of about 20 Meg Ohms, and even a tiny amount of contamination can appear on the computer screen as a significant load. Two cases are presented below:

1. Sensor models 6220 / 6230 lend themselves to being trimmed from the inside outward in order to accommodate sliding the sensor over a bolt or part diameter. To allow a fastener to pass through the center of the sensor, cut a hole in the center of the 6220 or 6230. The connecting traces for the spokes of this polar coordinate sensor are wired from the outside to the inside. The rings have connecting traces coming through the pie-shaped region that is not active. Thus, the inside of these sensors can be removed and the outside region will not be affected.

    - With a very sharp knife, or a hole-punch, cut the center out of the sensor, with a diameter 0.050” to 0.100” larger than the clearance diameter required.

    - Laminate thin tape over both sides of the sensor so that in the middle, where there is a cut hole, the adhesive from the tape on one side seals against the adhesive of the tape from the other side. It is desirable to use tape at least as wide as the active region of the sensor. This prevents either gaps in the tape coverage that would be thinner, or overlaps that would be thicker.

    - Cut a clearance hole through the center, making sure to leave a complete ring of tape around the edge, to maintain a seal for the cut edge. Either a very sharp knife, or a smaller hole-punch can be used for the clearance hole.

2. The 5250 sensor can be cut from 10” x 10” to a smaller size, such as 5” x 10”. Note that the connecting traces for the 5250 are routed on two sides of the active region. The side opposite the edge with connecting traces can be cut, and operation of the remaining portion of the sensor is not affected.

    - With a very sharp knife, trim the edge to be removed.

    - Attach half the width of a piece of tape to one side of the cut edge. Wrap the tape around the cut edge, and attach it to the other side, sealing the cut edge.

    - Since the region with the sealing tape is thicker than the sensor alone, the portion wrapped with tape should not be inserted into regions with hard surfaces coming together, unless tape or shim stock is placed on the rest of the sensor.

Shipping Cover on the Sensor

Most Tekscan sensors are constructed from two 0.001” mylar substrates and have a total thickness (where silver traces cross) of approximately 0.00385”. Many sensors are also shipped with a 0.003” mylar protective layer used to make handling easier in the manufacturing process. It also protects the sensor from becoming wrinkled in storage. Most users remove the cover before taking measurements. They prefer a thin, minimally intrusive sensor. Other users leave the cover in place to obtain added sensor life due to the extra protection it offers. When the cover is used in the experiment, it is recommended that the calibration include the protective cover as well.

- The sensor and cover together are thicker than the sensor alone. The added thickness may disturb measurement conditions.

- The cover has both an adhesive to hold it in place, and a release agent. When higher pressures are applied, the adhesive may become “set” in places, inducing squirm or shear effects on the surface of the sensor. When pressure is released, shadows of the applied load may remain on the sensor.

- The tab end of the sensor has a narrow rim of shipping cover that may delaminate inside the handle. When the lever is released to remove the sensor from the handle, the rim may “hang up” or “catch” on pins inside the handle, which in turn may damage the sensor or pins within the handle. If the pins lose alignment to the pads on the sensor, one or more contacts may be lost, and repair will be required to restore operation.

  If the sensor does not slide out easily, re-insert it and wiggle it gently back and forth in an attempt to unhook the cover from the pins. Gently try extracting the sensor once again.

  Peeling the proximal end of the shipping cover from the sensor tab before inserting it into the handle may avoid this problem. Additionally, the shipping cover can be peeled back and cut below the neck at the proximal end of the sensor tab.

The following image displays the shipping cover being peeled back from the sensor.

Removing the Shipping Cover

The first step to removing the shipping cover from the sensor is to start the de-lamination. Many array sensors are shipped with a small piece of paper between the sensor and the cover. This facilitates the start of de-lamination. For sensors without an inserted piece of paper, there are two suggestions for starting the process. Find a corner of the sensor, and rub (fan) the edge with a piece of adhesive, such as a piece of masking tape backed up by a soft rubber eraser. Alternately, carefully poke at the edge of a corner with a razor or Exacto knife to insert the blade between the sensor and the cover. If you choose this technique, please ensure that the knife is between the cover and sensor, and not between the top and bottom of the sensor.

When de-lamination has started, lay the sensor on a flat surface, such as a table. Now SLOWLY peel the cover back, while keeping the sensor as flat as possible on the table. The goal is to minimize air which could be induced into the sensor if it were peeled while being held in the air. Holding the sensor flat minimizes the possibility that one of the silver traces may be over-stressed and cracked as the cover is removed.

Cleaning the Sensor

Clean the sensor after each use by wiping it down with a damp rag or cloth with some alcohol. Store the sensors safely and flat (horizontal for large sensor pads) at room temperature in a protective case. Sensors that become wrinkled during storage may show artifacts during measurements. If the tab becomes wrinkled, continuity between handle pins and sensor pads may be lost.

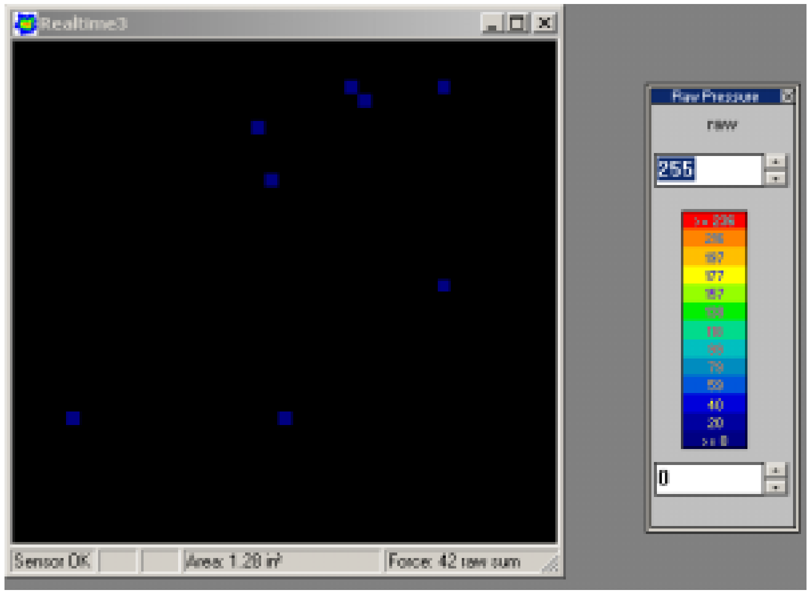
SOFTWARE CONSIDERATIONS

Reducing Random Noise

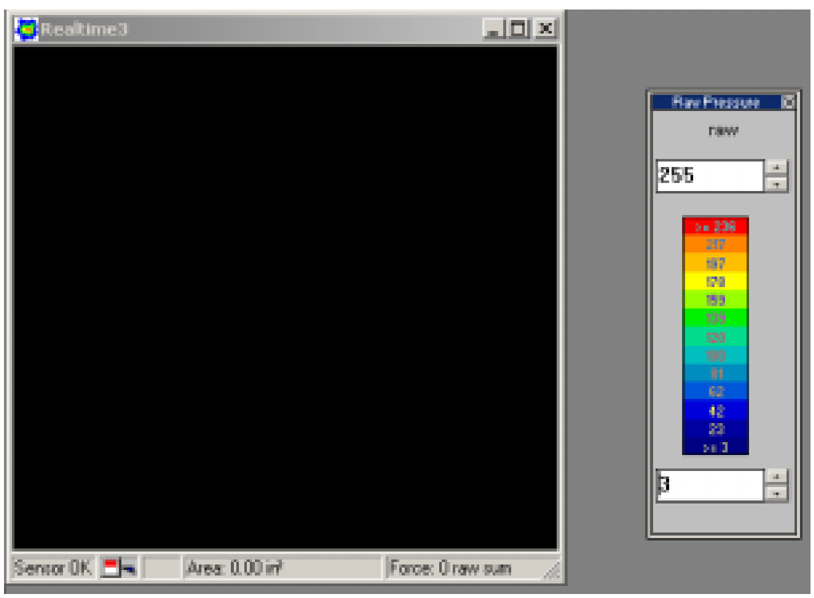
If there is random low-level noise, consider raising the lower limit on the legend. Adjusting the lower value on the legend will suppress noise and reduce artifacts. Adjustments to the legend change the displayed image, but do not affect the data stored in the movie.

Though raising the noise threshold will reduce random noise and artifact, rarely is it a good idea to perform this operation. Once the noise threshold is set, data from sensels with lower digital output is absent from the movie, and can never be recovered. If it is necessary, you can adjust the noise threshold through **Options → Aquisition Parameters**. The default threshold level (set at the factory) is 3, and this setting should be used for almost all measurements. This threshold tells the electronics in the handle to eliminate output from sensels with a raw digital output value of 3 or lower.

The following shows some Random noise with a full-scale Raw Digital Output set for 0-255 through the legend. Notice there are random artifacts.



The following shows the same sensor image. Notice that with the legend set for 3-255 there are no random artifacts.



Calibration Files

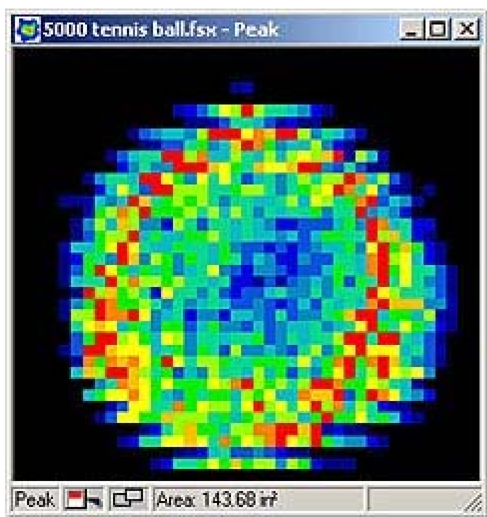
You may generate and save many calibration files for a given sensor and load those files in the future as long as you apply the calibration files to that same physical sensor with which you calibrated and recorded. Many users identify each sensor by marking it with a felt-tip pen: A, B, C, etc. Then they use that identity in the name of the calibration file. Similarly, many users include the calibration force in the name of the calibration file. For example, **Sensor B 100 lb.**

Viewing the Equilibration Process

When equilibrating, use the 2-D view to clearly see each sensel as a square pixel on your computer screen. If Avg1 (Averaging) or Avg2 (Fixed Area Averaging) is turned on, the granularity of the view is reduced (the view is smoothed out), and it is not possible to distinguish each sensel.

The 2-D view displays the recording in two-dimensional form with the color of each sensel location representing the pressure sensed at that point. This display looks the closest to the actual raw output of the sensor, and individual sensels can be identified. **2-D** is the default view.

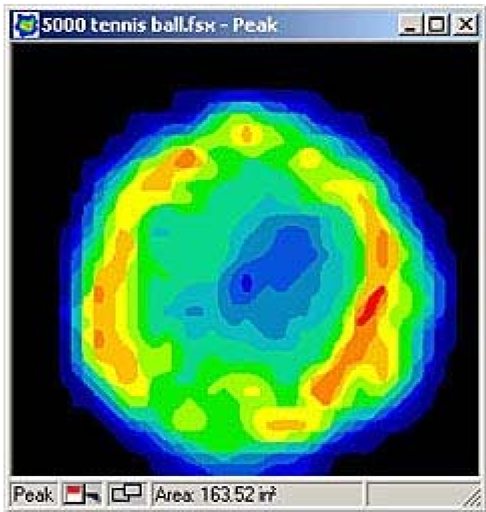
The following displays the 2-D view without Avg1 (Averaging) or Avg2 (Fixed Area Averaging) turned on:

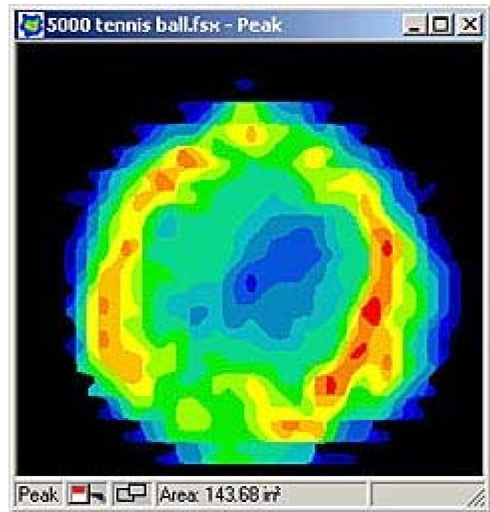


The following 2 images demonstrates what the sensor looks like with Avg1 or Avg2 turned on:

Averaging 

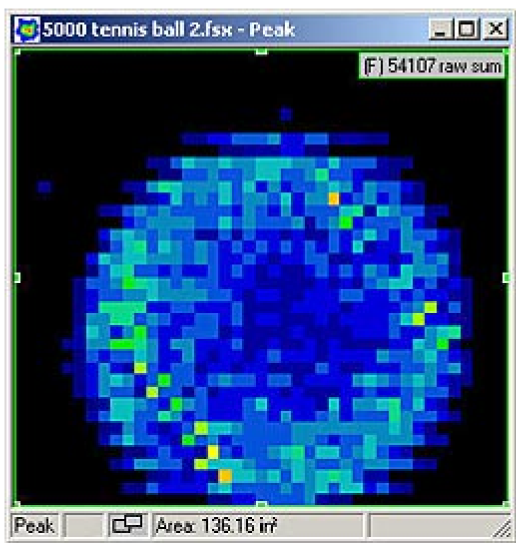
Fixed Area Averaging 



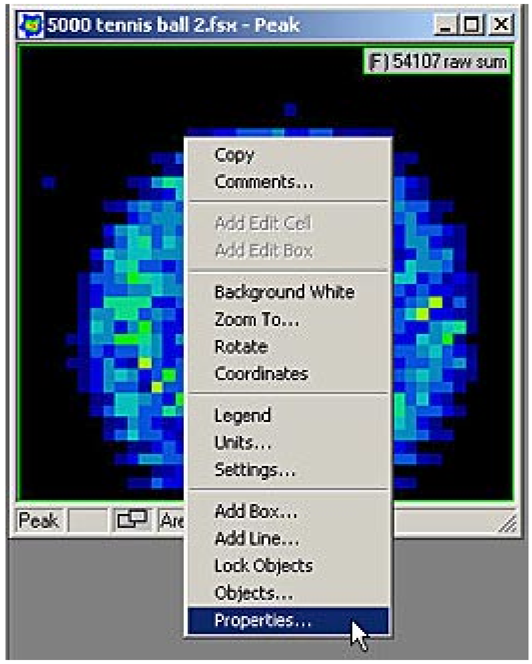


To follow the effects of equilibration, do the following:

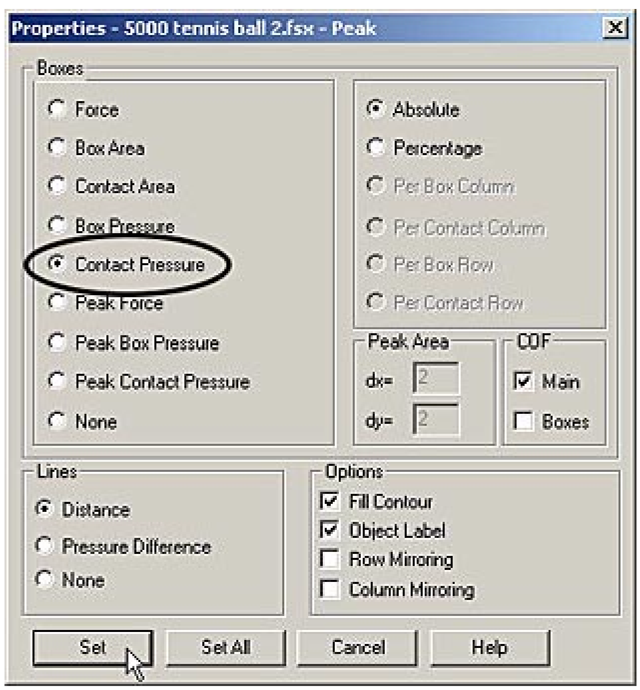
1. place an analysis box over the entire sensor surface.



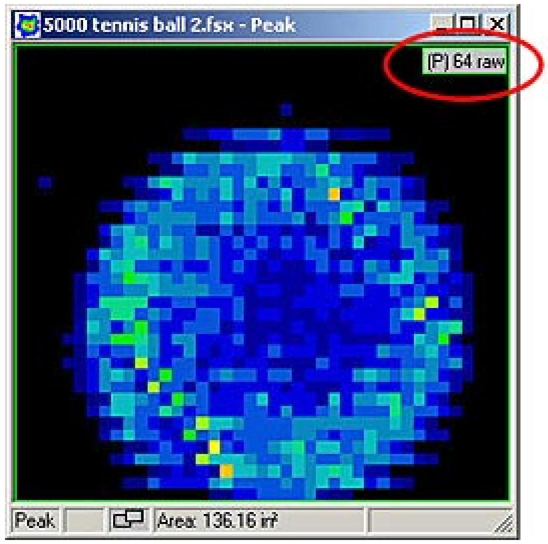
2. Right-click on the sensor image, and then click on **Properties** (the bottom command in the menu list).



3. For box properties, select "Contact Pressure" and then click on the **Set** button.

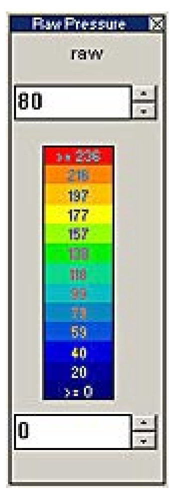


This shows the average contact pressure on the sensor in Raw Digital Output.



4. In the equilibration fixture, apply the first equilibration pressure. Many sensors display trapped air when first put into a uniform pressure field. Trapped air appears as a region of low reported pressure that results from an air bubble. Over time, the trapped air will dissipate. It may be advantageous to apply a pressure considerably greater than the equilibration pressure, and wait while the trapped air bleeds out. Then, without allowing the pressure in the bladder to go to zero, reduce the applied pressure to the first equilibration point. *Refer to "Calibration Load" and "Low Pressure Application" for more information on Trapped Air in the sensor.*

5. From within the software, click on the legend icon on the toolbar. The legend's units should be set for Raw Digital Output. Initially, the upper and lower limits will be set at the value from the previous time the legend was utilized. Now, adjust the upper and lower limits on the legend to bracket the average Raw Digital Output displayed on the upper right of the box. For example, if a six-point equilibration is to be done at average raw counts of 40, 80, 120, 160, 200, and 240, the lower and upper values could be set to the last and next equilibration values respectively. Thus, for equilibration at a raw count of 40, set the legend's lower limit to 0 and upper limit to 80.



Comparing Equilibrated and Non-Equilibrated Data

To compare the effect of equilibration with non-equilibrated data, record a movie with equilibration invoked. Open this movie in ***I-Scan*** twice. Then click on one of the displayed screen images to make it the active window. Go into **Tools → Equilibration**. When the equilibration window appears, it should show the equilibration array in shades of gray, and in the upper left corner of the window there should be a checkmark in front of the words “Equilibration Enabled.” Clicking on this checkmark will remove equilibration from the active movie. Now, the two versions of the same movie (one with equilibrated data and the other with non-equilibrated data) can be played back side-by-side. Analysis boxes can be placed on both versions to compare and graph data such as peak pressure.

Note

These comments are intended to assist you to achieve your application goals. If these suggestions do not resolve your needs, please contact Tekscan customer support. To assist the discussion, please send some frames or movies that illustrate the issues. Please also include a comment or two about the nature of the surfaces that come together, forces, pressures, and load duration involved. Tekscan is not seeking proprietary information about what you are doing. Troubleshooting is greatly improved by knowing the measurement context. This makes the circumstances clear and understandable.